Convergence Analysis of the Quantum Blahut-Arimoto Algorithm and the Contraction Coefficient of Qubit Amplitude Damping Channels

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Abstract

In this work, we investigate the convergence rate of the quantum Blahut–Arimoto (BA) algorithm proposed for computing the coherent information of less noisy quantum channels. Previous works established exponential convergence of the quantum Blahut-Arimoto algorithm under certain assumptions on the channel, but these assumptions are generally difficult to verify. In this work, using a Taylor expansion analysis around the optimal solution, we derive a more tractable criterion under which the gap between the optimal value and the value obtained at the n-th iteration decays exponentially with the number of iterations. In particular, we show that the condition is satisfied for qubit amplitude damping channels.

In addition, by analyzing the Bogoliubov–Kubo–Mori (BKM) metric on the Bloch ball, we derive an explicit characterization of the contraction coefficient of qubit amplitude damping channels. The proposed approach is further extended to calculate the relative contraction coefficient and the relative expansion coefficient of two qubit amplitude damping channels.

Keywords: Blahut-Arimoto algorithm, coherent information, amplitude damping channel, BKM metric, contraction coefficient.

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1 Introduction

In classical information theory, a problem of great importance is to compute the capacity of channels which characterize the highest rate of reliable communication. Since Shannon's foundational work [1], channel capacity has been a basic quantity in information theory. Unfortunately, capacity of most channels lack closed analytical expression so an optimization problem is given instead. Specifically, for a discrete memoryless channel P, the capacity is given by the formula

$$C(P) = \max_{p} I(p, P) \tag{1}$$

where p denotes the input distribution and $I(p, P)$ denotes the mutual information induced by the channel P under the input distribution p.

A fundamental numerical method for this problem is the Blahut-Arimoto algorithm that alternately updates the posterior distributions and the input distributions, independently developed by Arimoto and Blahut for computing channel capacitys and rate-distortion functions [2, 3]. The algorithm offers several significant advantages compared to gradient-based methods, such as its extremely fast convergence under specific conditions, which can even achieve an exponential rate. However, the exponential convergence result is not obtained from a global contraction condition. Rather, Arimoto's argument uses Taylor expansion to study the behavior around the capacity-achieving input distribution. This local viewpoint is the main motivation for our analysis of the quantum Blahut-Arimoto algorithm. Beyond its original formulation, the Blahut-Arimoto iteration has also been interpreted through information geometry and proximal-point methods, where the relative entropy plays the role of a natural divergence controlling the geometry of the update [4, 5]. These viewpoints help explain why the practical convergence behavior of the algorithm is often much better than what can be derived from crude worst-case bounds.

Quantum information theory, which lies at the intersection of information theory, operator theory, and quantum mechanics, has developed rapidly in recent decades. A central problem in quantum information theory is to determine the ultimate rate at which information can be reliably transmitted through a noisy quantum channel. Unlike the classical case, quantum channels give rise to several different capacity notions, depending on whether the transmitted information is classical or quantum and on whether additional resources such as pre-shared entanglement are available. As an example, the Holevo-Schumacher-Westmoreland theorem characterizes the classical communication rate of classical-quantum channels [6–8].

Among the entropic quantities appearing in quantum communication, coherent information plays a fundamental role in the study of quantum capacity. It measures the difference between the information received by the legitimate receiver and the information leaked to the environment through the complementary channel. Coherent information appears in the Lloyd-Shor-Devetak formula for quantum capacity and also plays a central role in the study of private communication and single-letter capacity formulas for special classes of channels [9–11]. Thus, coherent information is not merely a quantum analogue of classical mutual information. Rather, it lies at the intersection of coding theorems, degradability-type conditions, and less-noisy comparisons.

In general, computing coherent information is a difficult optimization problem over the set of quantum states. The difficulty comes from both the non-commutative structure of quantum states and the presence of matrix logarithms in the von-Neumann entropy. Even when the problem becomes more tractable for special classes of channels, efficient numerical methods with rigorous convergence guarantees remain highly desirable. Existing approaches include convex optimization methods, semi-definite approximations of the matrix logarithm, and geometric algorithms for special quantum-state spaces [12, 13].

The Blahut-Arimoto philosophy has also been extended to quantum information theory. Early quantum Blahut-Arimoto-type algorithms were studied in connection with quantum channel capacity computations and numerical tests of additivity questions [14, 15]. Ramakrishnan's work [16] generalized the classical Blahut–Arimoto algorithm to the quantum setting and provided a unified computational framework for computing various quantum capacities including the coherent information. Subsequent work further generalized these algorithms to broader quantum information bottleneck and variational settings [17]. A sufficient condition for exponential convergence of the quantum Blahut-Arimoto algorithm was formulated in terms of the expansion coefficient of the quantum channel. In particular, the algorithm converges exponentially fast when this coefficient is strictly positive. However, Belzig et al. show that, for non-unitary channels whose input dimension is larger than the output dimension, the expansion coefficient is zero [18]. This indicates that the global expansion-coefficient condition may be too restrictive for capturing the asymptotic convergence behavior of the algorithm, and motivates a more local analysis around the optimizer.

The key observation in our paper is that the proposed quantum Blahut-Arimoto algorithm converges to the optimal solution. Hence, after finitely many iterations, its behavior is determined by the local structure of the quantum channel near the optimum. Inspired by Arimoto's work, we employ Taylor expansion to analyze this local behavior and derive sufficient conditions for exponential convergence when computing the coherent information of quantum less noisy channels. Here, "local" means that once the number of iterations is sufficiently large, the convergence rate

becomes exponential.

As an application of this local criterion, we consider the qubit amplitude damping channel. This channel is one of the basic noise models in quantum information theory, describing energy dissipation from an excited state to a ground state. It is also a natural testing ground for algorithms that compute coherent information, because its structure is simple enough to allow detailed analysis while still retaining genuinely quantum behavior. We verify that qubit amplitude damping channels satisfies our sufficient conditions.

The second part of this paper concerns contraction coefficients of quantum channels with respect to quantum relative entropy. The data processing inequality asserts that quantum relative entropy is monotone nonincreasing under the action of quantum channels. The contraction coefficient refines this statement by asking for the best constant that quantifies how strongly a fixed channel decreases relative entropy. Thus, the contraction coefficient measures the worst-case loss of distinguishability caused by the channel. Such coefficients are closely related to strong data processing inequalities and provide quantitative information about the irreversible effect of noise.

Computing relative entropy contraction coefficients is generally challenging, because it requires optimizing over pairs of quantum states. A useful way to approach this problem is through the Bogoliubov–Kubo–Mori (BKM) metric. The BKM metric appears as the infinitesimal second-order form of quantum relative entropy [19]. Therefore, comparison of relative entropies can often be reduced to comparison of the corresponding BKM quadratic forms. This local geometric viewpoint is especially effective for qubit channels, where the Bloch-sphere representation allows the problem to be made more explicit. Motivated by this, we exploit the geometric action and symmetry properties of the qubit amplitude damping channel on the Bloch ball to reduce the original high-dimensional nonconvex optimization over pairs of quantum states to a single variable optimization problem.

We then extend the same method to compare two different qubit amplitude damping channels. More precisely, we compute the relative contraction coefficient between two such channels, which measures how much one amplitude damping channel contracts relative entropy compared with another. This relative comparison is finer than the contraction coefficient of a single channel and is naturally connected to reverse-type data processing inequalities and less-noisy comparisons between quantum channels.

The two main themes of this paper are connected by a common principle: the local geometry of quantum relative entropy controls both the convergence of the quantum Blahut–Arimoto algorithm and the contraction behavior of quantum channels. In the study of the Blahut–Arimoto algorithm, Taylor expansion near the optimizer reveals the local stability of the iteration and leads to exponential convergence. In the study of contraction coefficients, the BKM metric captures the infinitesimal behavior of relative entropy and allows the optimal contraction constant to be computed. Thus, although the two problems arise from different motivations, both are governed by the same local analytic structure.

The rest of the paper is structured as follows: Section II collects the necessary preliminaries for the subsequent analysis. Section III presents the main results of this paper and states the principal theorems. Section IV is devoted to the proof of the convergence results for the quantum Blahut-Arimoto algorithm. In particular, we verify them on several common qubit channels. Section V

contains the proofs of the results on relative entropy contraction coefficients.

2 Preliminary

In this section, we fix notations and recall several basic notions that will be used throughout the paper.

2.1 Quantum channels and the coherent information

Throughout this paper, all Hilbert spaces are assumed to be finite-dimensional. Unless otherwise specified, all logarithms are taken to base e. We denote Hilbert spaces by $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$, and so on. The space of linear operators from $\mathcal{X}$ to $\mathcal{Y}$ is denoted by $L(\mathcal{X}, \mathcal{Y})$, and we write $L(\mathcal{X})$ as a shorthand for $L(\mathcal{X}, \mathcal{X})$. The set of density operators on $\mathcal{X}$ is $D(\mathcal{X}) = \{\rho \in L(\mathcal{X}) : \rho \geq 0, \mathrm{Tr}(\rho) = 1\}$.The von Neumann entropy of a density operator $\rho \in D(\mathcal{X})$ is defined as

$$S(\rho) = -\mathrm{Tr}(\rho \log \rho). \tag{2}$$

We write $T(\mathcal{X}, \mathcal{Y})$ for the space of linear maps from $L(\mathcal{X})$ to $L(\mathcal{Y})$, and write $T(\mathcal{X}) = T(\mathcal{X}, \mathcal{X})$. A linear map $\Phi \in T(\mathcal{X}, \mathcal{Y})$ is called a *quantum channel* if it is completely positive and trace-preserving.

For two density operators ρ and σ, the *quantum relative entropy* is defined by

$$D(\rho\|\sigma) = \begin{cases} \mathrm{Tr}\left[\rho(\log \rho - \log \sigma)\right], & \text{if } \mathrm{supp}(\rho) \subseteq \mathrm{supp}(\sigma), \\ +\infty, & \text{otherwise.} \end{cases} \tag{3}$$

Equivalently, the support condition may be written as $\ker(\sigma) \subseteq \ker(\rho)$, and we denote it by $\rho \ll \sigma$.

For two density operators ρ and σ, the *quantum χ^2-divergence* is defined by

$$\chi^2(\rho\|\sigma) = \begin{cases} \mathrm{Tr}\left(\rho^2\sigma^{-1}\right) - 1, & \text{if } \mathrm{supp}(\rho) \subseteq \mathrm{supp}(\sigma), \\ +\infty, & \text{otherwise,} \end{cases} \tag{4}$$

where σ^{-1} is understood as the inverse on $\mathrm{supp}(\sigma)$. From [20], we have the the following inequality

$$D(\rho\|\sigma) \leq \log\left(1 + \chi^2(\rho\|\sigma)\right) \leq \chi^2(\rho\|\sigma). \tag{5}$$

Let $\Phi \in T(\mathcal{X}, \mathcal{Y})$ be a quantum channel. By the Stinespring dilation theorem, there exists a finite-dimensional environment system $\mathcal{E}$ and an isometry $U : \mathcal{X} \to \mathcal{Y} \otimes \mathcal{E}$ such that

$$\Phi(\rho) = \mathrm{Tr}_{\mathcal{E}}\left(U\rho U^*\right), \quad \rho \in D(\mathcal{X}), \tag{6}$$

where $\mathrm{Tr}_{\mathcal{E}}$ denotes the partial trace over the environment system $\mathcal{E}$. The *complementary channel* of Φ associated with this Stinespring representation is defined by

$$\Phi_c(\rho) = \mathrm{Tr}_{\mathcal{Y}}\left(U\rho U^*\right), \quad \rho \in D(\mathcal{X}). \tag{7}$$

A quantum channel Φ is said to be *less noisy* if and only if the private capacity of its complementary channel Φ_c is zero. The channel Φ is called *degradable* if there exists a quantum channel $\mathcal{D} : L(\mathcal{Y}) \to L(\mathcal{E})$ such that $\mathcal{D} \circ \Phi = \Phi_c$. In this case, $\mathcal{D}$ is called the degrading channel of Φ. It is well known that every degradable channel is less noisy.

The *coherent information* of Φ is defined by

$$I_{\mathrm{coh}}(\Phi) := \max_{\rho \in D(\mathcal{X})} I_{\mathrm{coh}}(\rho, \Phi), \tag{8}$$

where

$$I_{\mathrm{coh}}(\rho, \Phi) = S\left(\Phi(\rho)\right) - S\left(\Phi_c(\rho)\right) \tag{9}$$

denotes the coherent information of Φ with respect to input state ρ.

2.2 BKM metric and relative entropy coefficients

In this subsection, we recall the Bogoliubov-Kubo-Mori (BKM) metric and its connection with relative entropy coefficients. This will be the main tool for computing contraction coefficients and relative contraction coefficients later.

The *BKM metric* $g_\sigma(X)$ of a operator $X \in L(\mathcal{X})$ at density operator $\sigma \in D(\mathcal{X})$ is defined as

$$g_\sigma(X) = \begin{cases} \mathrm{Tr}\left(\int_0^\infty X^*(\sigma + rI)^{-1} X (\sigma + rI)^{-1} dr\right), & \text{if } \mathrm{supp}(X) \subseteq \mathrm{supp}(\sigma), \\ +\infty, & \text{otherwise.} \end{cases} \tag{10}$$

Let Φ be a quantum channel. The *contraction coefficient* $\eta(\Phi)$ and the *expansion coefficient* $\breve{\eta}(\Phi)$ of Φ with respect to quantum relative entropy are defined respectively by

$$\eta(\Phi) := \sup_{\substack{\rho \ll \sigma \\ \rho \neq \sigma}} \frac{D\left(\Phi(\rho) \| \Phi(\sigma)\right)}{D(\rho \| \sigma)} \quad \text{and} \quad \breve{\eta}(\Phi) := \inf_{\substack{\rho \ll \sigma \\ \rho \neq \sigma}} \frac{D\left(\Phi(\rho) \| \Phi(\sigma)\right)}{D(\rho \| \sigma)}. \tag{11}$$

Since $D(\rho \| \sigma) > 0$ whenever $\rho \neq \sigma$, the above ratios are well defined. By the data processing inequality, one always has

$$0 \leq \breve{\eta}(\Phi) \leq \eta(\Phi) \leq 1. \tag{12}$$

The corresponding coefficients with respect to the BKM metric are defined by taking the infinitesimal form of the above ratios. More precisely, the *BKM metric contraction coefficient* and the *BKM metric expansion coefficient* of Φ are defined respectively by

$$\eta_{\mathrm{BKM}}(\Phi) := \sup_{\substack{\sigma \in D(\mathcal{X}) \\ X = X^*,\ \mathrm{Tr}(X) = 0 \\ X \neq 0}} \frac{g_{\Phi(\sigma)}\left(\Phi(X)\right)}{g_\sigma(X)}, \tag{13}$$

and

$$\check{\eta}_{\mathrm{BKM}}(\Phi) := \inf_{\substack{\sigma\in D(\mathcal{X}) \\ X=X^*,\ \mathrm{Tr}(X)=0 \\ X\neq 0}} \frac{g_{\Phi(\sigma)}\left(\Phi(X)\right)}{g_{\sigma}(X)}. \tag{14}$$

Then we introduce relative versions of contraction and expansion coefficients. Let Φ and Ψ be two quantum channels with the same input system. The *relative contraction coefficient* and the *relative expansion coefficient* of Φ with respect to Ψ is defined respectively by

$$\eta(\Phi,\Psi) := \sup_{\substack{\rho\ll\sigma \\ \rho\neq\sigma}} \frac{D\left(\Phi(\rho)\|\Phi(\sigma)\right)}{D\left(\Psi(\rho)\|\Psi(\sigma)\right)} \tag{15}$$

and

$$\check{\eta}(\Phi,\Psi) := \inf_{\substack{\rho\ll\sigma \\ \rho\neq\sigma}} \frac{D\left(\Phi(\rho)\|\Phi(\sigma)\right)}{D\left(\Psi(\rho)\|\Psi(\sigma)\right)}. \tag{16}$$

Whenever both quantities are well defined, we have the reciprocal relation

$$\check{\eta}(\Phi,\Psi) = \frac{1}{\eta(\Psi,\Phi)}. \tag{17}$$

Similarly, the *BKM metric relative contraction coefficient* of Φ with respect to Ψ is defined by

$$\eta_{\mathrm{BKM}}(\Phi,\Psi) := \sup_{\substack{\sigma\in D(\mathcal{X}) \\ X=X^*,\ \mathrm{Tr}(X)=0 \\ X\neq 0}} \frac{g_{\Phi(\sigma)}\left(\Phi(X)\right)}{g_{\Psi(\sigma)}\left(\Psi(X)\right)}. \tag{18}$$

while the *BKM metric relative expansion coefficient* of Φ with respect to Ψ is defined by

$$\check{\eta}_{\mathrm{BKM}}(\Phi,\Psi) := \inf_{\substack{\sigma\in D(\mathcal{X}) \\ X=X^*,\ \mathrm{Tr}(X)=0 \\ X\neq 0}} \frac{g_{\Phi(\sigma)}\left(\Phi(X)\right)}{g_{\Psi(\sigma)}\left(\Psi(X)\right)}. \tag{19}$$

Whenever both quantities are well defined, we also have

$$\check{\eta}_{\mathrm{BKM}}(\Phi,\Psi) = \frac{1}{\eta_{\mathrm{BKM}}(\Psi,\Phi)}. \tag{20}$$

The following lemmas about BKM metric [18] are main tools for studying these coefficients.

Lemma 1 (Belzig *et al.*, Lemma IV.1)**.** *Let $\Phi : L(\mathcal{X}) \to L(\mathcal{Y})$ and $\Psi : L(\mathcal{X}) \to L(\mathcal{Z})$ be quantum channels. For constants $c_1, c_2 > 0$, the following two statements are equivalent:*

(1) For all density operators ρ and σ,

$$c_1 D\left(\Phi(\rho)\|\Phi(\sigma)\right) \leq D\left(\Psi(\rho)\|\Psi(\sigma)\right) \leq c_2 D\left(\Phi(\rho)\|\Phi(\sigma)\right). \tag{21}$$

(2) For all density operators ρ and traceless Hermitian operator X,

$$c_1 g_{\Phi(\rho)}\left(\Phi(X)\right) \leq g_{\Psi(\rho)}\left(\Psi(X)\right) \leq c_2 g_{\Phi(\rho)}\left(\Phi(X)\right). \tag{22}$$

As a consequence of Lemma 1, the relative entropy coefficients coincide with the corresponding BKM metric coefficients:

$$\eta(\Phi,\Psi)=\eta_{\mathrm{BKM}}(\Phi,\Psi),\quad \check{\eta}(\Phi,\Psi)=\check{\eta}_{\mathrm{BKM}}(\Phi,\Psi). \tag{23}$$

In particular, taking Ψ to be the identity channel gives

$$\eta(\Phi)=\eta_{\mathrm{BKM}}(\Phi),\quad \check{\eta}(\Phi)=\check{\eta}_{\mathrm{BKM}}(\Phi). \tag{24}$$

We now specialize to qubit systems. Let $\vec{\sigma}=(\sigma_x,\sigma_y,\sigma_z)$ denotes the vector of Pauli matrices. Every qubit density operator admits a Bloch representation of the form

$$\rho_{\vec{w}}=\frac{1}{2}(I_2+\vec{w}\cdot\vec{\sigma}),\quad |\vec{w}|\leq 1 \tag{25}$$

where $\vec{w}\in\mathbb{R}^3$ is the Bloch vector of $\rho_{\vec{w}}$. We shall also use the same Bloch coordinates for traceless Hermitian perturbations. More precisely, every traceless Hermitian operator can be written uniquely as

$$X_{\vec{v}}=\vec{v}\cdot\vec{\sigma},\quad \vec{v}\in\mathbb{R}^3. \tag{26}$$

For later computations, it is convenient to combine the above representation with the Bloch representation of qubit channels. If Φ is a qubit channel, then there exist a real matrix $T_\Phi\in M_3(\mathbb{R})$ and a vector $\vec{t_\Phi}\in\mathbb{R}^3$ such that

$$\Phi(\rho_{\vec{w}})=\frac{1}{2}\left(I_2+(T_\Phi\vec{w}+\vec{t_\Phi})\cdot\vec{\sigma}\right) \tag{27}$$

and

$$\Phi(X_{\vec{v}})=(T_\Phi\vec{v})\cdot\vec{\sigma}. \tag{28}$$

Let $f:[0,1]\to\mathbb{R}$ be defined by

$$f(x):=\frac{1-x^2}{2x}\log\frac{1+x}{1-x},\quad x\in(0,1], \tag{29}$$

with $f(0):=1$ by continuity. The following explicit formula [18] will be used later.

Lemma 2 (Belzig *et al.*, Lemma IV.4)**.** *Let $\rho_{\vec{w}}=\frac{1}{2}\left(I_2+\vec{w}\cdot\vec{\sigma}\right)$ and $X_{\vec{v}}=\vec{v}\cdot\vec{\sigma}$ satisfying $|\vec{w}|<1$. Then*

$$g_{\vec{w}}(\vec{v})\triangleq g_{\rho_{\vec{w}}}(X_{\vec{v}}) \tag{30}$$

$$=\frac{4|\vec{v}|^2}{1-|\vec{w}|^2}\left[\frac{|\vec{w}\cdot\vec{v}|^2}{|\vec{w}|^2|\vec{v}|^2}\left(1-f(|\vec{w}|)\right)+f(|\vec{w}|)\right]. \tag{31}$$

Let $\vec{w_\Phi}=T_\Phi\vec{w}+\vec{t_\Phi}$ and $\vec{y_\Phi}=T_\Phi\vec{v}$. Then

$$g_{\vec{w}_\Phi}(\vec{v}_\Phi)\triangleq g_{\Phi(\rho_{\vec{w}})}\left(\Phi(X_{\vec{v}})\right) \tag{32}$$

$$=\frac{4|\vec{v}_\Phi|^2}{1-|\vec{w}_\Phi|^2}\left[\frac{|\vec{w}_\Phi\cdot\vec{v}_\Phi|^2}{|\vec{w}_\Phi|^2|\vec{v}_\Phi|^2}\left(1-f(|\vec{w}_\Phi|)\right)+f(|\vec{w}_\Phi|)\right]. \tag{33}$$

2.3 Structure of the quantum Blahut-Arimoto algorithm

In this subsection, we recall the quantum Blahut-Arimoto algorithm introduced in [16], which is used to compute the coherent information of less noisy channels.

Let Φ be a less noisy channel. Define

$$F_{\Phi}(\rho) := \Phi_c^* \log \Phi_c(\rho) - \Phi^* \log \Phi(\rho). \tag{34}$$

For density operators $\rho, \sigma \in D(\mathcal{X})$ with $\rho \ll \sigma$, let

$$J_\gamma(\rho, \sigma, \Phi) := -\gamma D(\rho \| \sigma) + \mathrm{Tr}\left[\rho F_{\Phi}(\sigma)\right], \tag{35}$$

where $\gamma > 0$ is an acceleration parameter.

For a less noisy channel Φ, one has

$$D\left(\Phi(\rho) \| \Phi(\sigma)\right) - D\left(\Phi_c(\rho) \| \Phi_c(\sigma)\right) \geq 0 \tag{36}$$

for all admissible pairs ρ, σ. We choose γ such that

$$\sup_{\substack{\rho \ll \sigma \\ \rho \neq \sigma}} \frac{D\left(\Phi(\rho) \| \Phi(\sigma)\right) - D\left(\Phi_c(\rho) \| \Phi_c(\sigma)\right)}{D(\rho \| \sigma)} \leq \gamma \leq 1. \tag{37}$$

With this choice, it is easy to verify that $J_\gamma(\rho, \sigma, \Phi) \leq J_\gamma(\rho, \rho, \Phi)$ and therefore

$$I_{\mathrm{coh}}(\Phi) = \max_{\rho \in D(\mathcal{X})} \max_{\sigma \in D(\mathcal{X}): \rho \ll \sigma} J_\gamma(\rho, \sigma, \Phi) \tag{38}$$

$$= \max_{\rho \in D(\mathcal{X})} J_\gamma(\rho, \rho, \Phi). \tag{39}$$

The algorithm proposed in [16] consists of the following steps:

(1) Initially, choose a full-rank density operator $\rho^{(1)} \in D(\mathcal{X})$. In practice, the maximally mixed state $\frac{id_{\mathcal{X}}}{\dim(\mathcal{X})}$ is a natural choice.

(2) For $t \in \mathbb{N}^+$, update

$$\rho^{(t+1)} = \frac{1}{Z(t+1)} \exp\left[\log \rho^{(t)} + \frac{1}{\gamma} F_{\Phi}\left(\rho^{(t)}\right)\right], \tag{40}$$

where the normalization constant is given by

$$Z(t+1) = \mathrm{Tr}\left\{\exp\left[\log \rho^{(t)} + \frac{1}{\gamma} F_{\Phi}(\rho^{(t)})\right]\right\}. \tag{41}$$

(3) The value generated by the algorithm at the t-th step is

$$I_{\mathrm{coh}}^{(t)}(\Phi) := J_\gamma(\rho^{(t+1)}, \rho^{(t)}, \Phi). \tag{42}$$

Let ρ^* be an optimizer of the coherent information, that is,

$$\rho^* \in \arg\max_{\rho \in D(\mathcal{X})} I_{\mathrm{coh}}(\rho, \Phi). \tag{43}$$

Ramakrishnan et al. prove the following properties, which will be used later.

(1) The sequence $I_{\mathrm{coh}}^{(t)}(\Phi)$ is monotonically increasing in t and converges to $I_{\mathrm{coh}}(\Phi)$.

(2) For every $t \in \mathbb{N}^+$,

$$0 \leq I_{\mathrm{coh}}(\Phi) - I_{\mathrm{coh}}^{(t)}(\Phi) \leq \gamma\, D(\rho^* \| \rho^{(t)}). \tag{44}$$

The convergence of the quantum Blahut-Arimoto algorithm has also been studied by Ramakrishnan et al. In particular, they prove the following global exponential convergence result.

Proposition 1 (Ramakrishnan *et al.*, Proposition III.6)**.** *Let $\left\{\rho^{(t)}\right\}_{t\geq 1}$ be the sequence generated by the quantum Blahut-Arimoto iteration for computing the coherent information of a less noisy channel Φ. Suppose that there exists a constant $a \in (0, \gamma]$ such that*

$$\inf_{\rho \neq \sigma} \frac{D\left(\Phi(\rho) \| \Phi(\sigma)\right) - D\left(\Phi_c(\rho) \| \Phi_c(\sigma)\right)}{D(\rho \| \sigma)} = a. \tag{45}$$

Then the optimizer ρ^* is unique and

$$I_{\mathrm{coh}}(\Phi) - I_{\mathrm{coh}}^{(t+1)}(\Phi) \leq \gamma \left(1 - \frac{a}{\gamma}\right)^t D(\rho^* \| \rho^{(1)}). \tag{46}$$

The quantity

$$\inf_{\rho \neq \sigma} \frac{D\left(\Phi(\rho) \| \Phi(\sigma)\right) - D\left(\Phi_c(\rho) \| \Phi_c(\sigma)\right)}{D(\rho \| \sigma)} \tag{47}$$

is bounded above by the expansion coefficient of Φ. However, recent results on reverse data processing inequalities indicate that such a global positivity condition is extremely restrictive [18]. In particular, for a quantum channel whose input dimension is not smaller than its output dimension, the ordinary relative expansion coefficient must vanish unless the channel is unitary. Consequently, one should not expect assumptions of Proposition 1 to hold for generic quantum channels.

This observation motivates the main idea of our work. Rather than seeking a global expansion coefficient that is uniformly positive over the entire state space, we investigate whether a local expansion phenomenon emerges along the Blahut-Arimoto trajectory. The key point is that, although the global infimum may be zero, the proof of Proposition 1 only relies on the one-step estimate

$$D\left(\Phi(\rho^*) \| \Phi\left(\rho^{(t)}\right)\right) - D\left(\Phi_c(\rho^*) \| \Phi_c(\rho^{(t)})\right) \geq a\, D(\rho^* \| \rho^{(t)}). \tag{48}$$

Whenever such an inequality can be established locally around the optimizer, one still obtains a contraction estimate for $D(\rho^* \| \rho^{(t)})$ which serves as the starting point of our local convergence

analysis. More specifically, one has that the relative entropy distance to the optimizer contracts at the next step:

$$D(\rho^*\|\rho^{(t+1)}) \leq \left(1 - \frac{a}{\gamma}\right) D(\rho^*\|\rho^{(t)}) \tag{49}$$

whenever (48) holds.

3 Main Results

In this section we state our main results. The first result, presented in Subsection 3.1, concerns the local convergence behavior of the quantum Blahut-Arimoto algorithm for computing the coherent information. The second result, presented in Subsection 3.2, concerns amplitude damping channels: we characterize the contraction coefficient of a single amplitude damping channel, as well as the relative contraction coefficient between two amplitude damping channels.

3.1 Exponential convergence of the quantum Blahut-Arimoto algorithm

In this subsection we establish sufficient conditions under which the quantum Blahut-Arimoto algorithm converges locally at an exponential rate. We then verify that these conditions are satisfied by the qubit dephasing channel and the qubit amplitude damping channel in appropriate parameter regimes.

We first state the main convergence theorem. The assumptions below imply the strict concavity of the coherent information, and hence the uniqueness of the optimizer. Under the additional assumption that the optimizer is positive-definite, these assumptions further yield the local exponential convergence of the quantum Blahut-Arimoto algorithm.

Theorem 1. *Let $\mathcal{X}$ and $\mathcal{Y}$ be finite-dimensional Hilbert spaces of dimensions m and n, respectively. Let $\Phi \in T(\mathcal{X}, \mathcal{Y})$ be a degradable quantum channel with the degrading channel $\mathcal{D}$. Assume that Φ is injective and $\eta(\mathcal{D}) < 1$. Then $I_{\mathrm{coh}}(\rho, \Phi)$ is strictly concave with respect to ρ. Consequently, the optimizer ρ^* of $I_{\mathrm{coh}}(\Phi)$ is unique.*

Assume further that ρ^* is positive-definite. Let $\left\{\rho^{(t)}\right\}_{t\in\mathbb{N}^+}$ be the sequence generated by the quantum Blahut-Arimoto algorithm with parameter γ. Then there exist $N_0 \in \mathbb{N}^+$ and a constant $a \in (0, \gamma]$ such that, for all $t \geq N_0$,

$$I_{\mathrm{coh}}(\Phi) - I_{\mathrm{coh}}^{(t)}(\Phi) \leq \gamma \left(1 - \frac{a}{\gamma}\right)^{t-N_0} D\left(\rho^*\|\rho^{(N_0)}\right). \tag{50}$$

Then we verify the above assumptions for two important families of qubit channels. We use the following standard parametrizations.

The qubit dephasing channel with parameter $p \in (0, 1)$ is defined by

$$\Psi_p(\rho) = (1-p)\rho + pZ\rho Z, \tag{51}$$

where $Z = |0\rangle\langle 0| - |1\rangle\langle 1|$ is the Pauli-Z operator.

The qubit amplitude damping channel with parameter $p \in [0,1]$ is defined by

$$\Phi_p(\rho) = A_0 \rho A_0^* + A_1 \rho A_1^*, \tag{52}$$

where $A_0 = |0\rangle\langle 0| + \sqrt{1-p}\,|1\rangle\langle 1|$ and $A_1 = \sqrt{p}\,|0\rangle\langle 1|$.

With the above parametrizations, we are now ready to state the following corollary.

Corollary 1. *The qubit dephasing channel with parameter $p \in (0, \frac{1}{2}) \cup (\frac{1}{2}, 1)$ and the qubit amplitude damping channel with parameter $p \in [0, \frac{1}{2})$ are injective and degradable. Moreover, the contraction coefficients of their respective degrading channels are strictly less than 1.*

3.2 Contraction coefficients for amplitude damping channels

In this subsection we first sharpen the existing estimate

$$\eta(\Phi_p) \leq \sqrt{1-p} \tag{53}$$

for the contraction coefficient of a single qubit amplitude damping channel [18]. Second, they give explicit formulas for the relative contraction and expansion coefficients between two amplitude damping channels.

We adopt the convention $\operatorname{arctanh}(x) = \frac{1}{2}\log\frac{1+x}{1-x}$. Our first theorem gives an exact one-dimensional variational formula for the contraction coefficient of Φ_p. The endpoint cases are trivial: $\eta(\Phi_0) = 1$ and $\eta(\Phi_1) = 0$.

Theorem 2. *For $p \in (0,1)$, the contraction coefficient of Φ_p with respect to the quantum relative entropy is given by*

$$\eta(\Phi_p) = \sup_{r \in [0,1)} (1-p) \frac{r}{\operatorname{arctanh}(r)} \frac{\operatorname{arctanh}\left(p + (1-p)r\right)}{p + (1-p)r}. \tag{54}$$

Here the expression at $r = 0$ is understood by taking the limit.

Next, we consider the comparison between two amplitude damping channels. Let $0 < p_1 < p_2 < 1$ and $q = \frac{p_2 - p_1}{1 - p_1}$. By the composition rule for amplitude damping channels

$$\Phi_{p_2} = \Phi_q \circ \Phi_{p_1}, \tag{55}$$

the channel Φ_{p_2} may be viewed as the result of applying an additional amplitude damping channel Φ_q after Φ_{p_1}. The following theorem gives the corresponding relative contraction coefficient.

Theorem 3. *For $0 < p_1 < p_2 < 1$ and $q = \frac{p_2 - p_1}{1 - p_1}$, we have*

$$\eta\left(\Phi_{p_2}, \Phi_{p_1}\right) = \sup_{r \in I_{p_1}} (1-q) \frac{r}{\operatorname{arctanh}(r)} \frac{\operatorname{arctanh}\left(q + (1-q)r\right)}{q + (1-q)r}, \tag{56}$$

where $I_{p_1} = \begin{cases} [0,1), & p_1 \in \left(0, \frac{1}{2}\right], \\ [2p_1 - 1, 1), & p_1 \in \left(\frac{1}{2}, 1\right). \end{cases}$ The value of the supremum at $r = 0$ is understood in the limiting sense.

Finally, we consider the reverse comparison, namely $\eta\left(\Phi_{p_1}, \Phi_{p_2}\right)$ for $0 < p_1 < p_2 < 1$. This is equivalent to computing the relative expansion coefficient $\check{\eta}\left(\Phi_{p_2}, \Phi_{p_1}\right)$. The following theorem gives its exact value.

Theorem 4. *For $0 < p_1 < p_2 < 1$, we have*

$$\check{\eta}\left(\Phi_{p_2}, \Phi_{p_1}\right) = \frac{p_1(1-p_2)}{p_2(1-p_1)}. \tag{57}$$

4 Proof of the exponential convergence of the quantum Blahut-Arimoto algorithm

4.1 Proof of Theorem 1

Proof. Arbitrarily choose an ensemble $\{p_i, \rho_i\}$ and let $\rho = \sum_i p_i \rho_i$. Then we have that

$$I_{\mathrm{coh}}\left(\sum_i p_i\rho_i, \Phi\right) - \sum_i p_i I_{\mathrm{coh}}\left(\rho_i, \Phi\right) \tag{58}$$

$$= S\left(\Phi(\rho)\right) - S\left(\Phi_c(\rho)\right) - \sum_i p_i S\left(\Phi(\rho_i)\right) + \sum_i p_i S\left(\Phi_c(\rho_i)\right) \tag{59}$$

$$= \left[S\left(\Phi(\rho)\right) - \sum_i p_i S\left(\Phi(\rho_i)\right)\right] - \left[S\left(\Phi_c(\rho)\right) - \sum_i p_i S\left(\Phi_c(\rho_i)\right)\right] \tag{60}$$

$$= \sum_i p_i D\left(\Phi(\rho_i)\|\Phi(\rho)\right) - \sum_i p_i D\left(\Phi_c(\rho_i)\|\Phi_c(\rho)\right) \tag{61}$$

$$= \sum_i p_i \left[D\left(\Phi(\rho_i)\|\Phi(\rho)\right) - D\left(\mathcal{D}\circ\Phi(\rho_i)\|\mathcal{D}\circ\Phi(\rho)\right)\right] \tag{62}$$

$$\geq \sum_i p_i \left(1-\eta(\mathcal{D})\right) D\left(\Phi(\rho_i)\|\Phi(\rho)\right), \tag{63}$$

where (62) follows from the assumption that Φ is degradable and (63) follows from the strong data processing inequality for the degrading channel $\mathcal{D}$. From the injectivity of Φ we have that $D\left(\Phi(\rho_i)\|\Phi(\rho)\right) > 0$. From above calculation, we can derive that $I_{\mathrm{coh}}(\rho, \Phi)$ is strictly concave with respect to ρ which implies ρ^* is unique.

Indeed, $J_\gamma(\rho, \rho, \Phi)$ is continuous [16] with respect to ρ, and $J_\gamma(\rho^{(t)}, \rho^{(t)}, \Phi)$ converges to

$$I_{\mathrm{coh}}(\Phi) = \max_\rho J_\gamma(\rho, \rho, \Phi) = J_\gamma(\rho^*, \rho^*, \Phi). \tag{64}$$

Therefore, $\rho^{(t)}$ converges to ρ^*, i.e. $\|\rho^{(t)} - \rho^*\| \to 0$ as $t \to +\infty$. Hence there is some $t_0 > 0$ such that $\rho^{(t)} > 0$ for all $t \geq t_0$.

Let $\rho^{(t)} = \rho^* + r_t H_t$ where H_t is traceless Hermitian operators with $\|H_t\| = 1$. Define

$$f_t(x) = \mathrm{Tr}\left[(\rho^*)^2 \left(\rho^* + xH_t\right)^{-1}\right] - 1. \tag{65}$$

Then we have

$$f_t'(x) = \mathrm{Tr}\left[(\rho^*)^2 \frac{d}{dx}(\rho^* + xH_t)^{-1}\right] \tag{66}$$

$$= -\mathrm{Tr}\left[(\rho^*)^2(\rho^* + xH_t)^{-1}H_t(\rho^* + xH_t)^{-1}\right] \tag{67}$$

and

$$f_t''(x) = 2\mathrm{Tr}\left[(\rho^*)^2(\rho^* + xH_t)^{-1}H_t(\rho^* + xH_t)^{-1}H_t(\rho^* + xH_t)^{-1}\right]. \tag{68}$$

Suppose $(\rho^*)^{-1}$ has the spectral decomposition $(\rho^*)^{-1} = \sum_{i=1}^{n} \lambda_i |y_i\rangle \langle y_i|$ $(0 < \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n)$ while $\rho^* - \rho^{(t)}$ has the spectral decomposition $\rho^* - \rho^{(t)} = \sum_{j=1}^{n} \nu_j |x_j\rangle \langle x_j|$. Then we have

$$\langle \rho^* - \rho^{(t)}, (\rho^*)^{-1}(\rho^* - \rho^{(t)})\rangle = \sum_{i,j,k=1}^{n} \nu_i \langle x_i|y_j\rangle \langle y_j|x_k\rangle \lambda_j \nu_k \langle x_k|x_i\rangle \tag{69}$$

$$= \sum_{i,j=1}^{n} {\nu_i}^2 \lambda_j |\langle x_i|y_j\rangle|^2 \tag{70}$$

$$\leq \lambda_n \sum_{i,j=1}^{n} {\nu_i}^2 \langle x_i|y_j\rangle \langle y_j|x_i\rangle \tag{71}$$

$$= \lambda_n \langle \rho^* - \rho^{(t)}, \rho^* - \rho^{(t)}\rangle. \tag{72}$$

Hence, there exists an integer $N_0 \in \mathbb{N}$ such that

$$f_t(r_t) = \chi^2(\rho^* \| \rho^{(t)}) \tag{73}$$

$$= (r_t)^2 \mathrm{Tr}\left(H_t(\rho^*)^{-1}H_t\right) + O\left(\|\rho^* - \rho^{(t)}\|^3\right) \tag{74}$$

$$= \langle \rho^* - \rho^{(t)}, (\rho^*)^{-1}(\rho^* - \rho^{(t)})\rangle + O\left(\|\rho^* - \rho^{(t)}\|^3\right) \tag{75}$$

$$\leq 2\lambda_n \langle \rho^* - \rho^{(t)}, \rho^* - \rho^{(t)}\rangle \tag{76}$$

for all $t \geq N_0$.

We equip $L(\mathcal{X})$ and $L(\mathcal{Y})$ with the Hilbert-Schmidt inner product $\langle A, B\rangle = \mathrm{Tr}(A^*B)$. Then $L(\mathcal{X})$ is a finite-dimensional Hilbert space, and $\Phi^*\Phi$ is a self-adjoint linear operator on $L(\mathcal{X})$. For every $X \in L(\mathcal{X})$, we have

$$\langle X, \Phi^*\Phi(X)\rangle = \langle \Phi(X), \Phi(X)\rangle \geq 0. \tag{77}$$

Moreover, equality holds if and only if $\Phi(X) = 0$, which, by the injectivity of Φ, is equivalent to $X = 0$. Thus $\Phi^*\Phi$ is positive definite. Therefore its smallest eigenvalue satisfies $\mu_{\min} := \lambda_{\min}(\Phi^*\Phi) > 0$.

In conclusion, we derive that

$$\begin{aligned}
&\frac{D\left(\Phi\left(\rho^*\right)\|\Phi\left(\rho^{(t)}\right)\right)-D\left(\Phi_c\left(\rho^*\right)\|\Phi_c\left(\rho^{(t)}\right)\right)}{D\left(\rho^*\|\rho^{(t)}\right)} \\
&\geq (1-\eta(\mathcal{D}))\,\frac{D\left(\Phi\left(\rho^*\right)\|\Phi\left(\rho^{(t)}\right)\right)}{D\left(\rho^*\|\rho^{(t)}\right)} &(78)\\
&\geq (1-\eta(\mathcal{D}))\,\frac{D\left(\Phi\left(\rho^*\right)\|\Phi\left(\rho^{(t)}\right)\right)}{\chi\left(\rho^*\|\rho^{(t)}\right)} &(79)\\
&\geq \frac{1-\eta(\mathcal{D})}{4\lambda_n}\,\frac{\langle\Phi\left(\rho^*-\rho^{(t)}\right),\Phi\left(\rho^*-\rho^{(t)}\right)\rangle}{\langle\rho^*-\rho^{(t)},\rho^*-\rho^{(t)}\rangle} &(80)\\
&= \frac{1-\eta(\mathcal{D})}{4\lambda_n}\,\frac{\langle\rho^*-\rho^{(t)},\Phi^*\Phi\left(\rho^*-\rho^{(t)}\right)\rangle}{\langle\rho^*-\rho^{(t)},\rho^*-\rho^{(t)}\rangle} &(81)\\
&\geq \frac{1-\eta(\mathcal{D})}{4\lambda_n}\,\mu_{min}, &(82)
\end{aligned}$$

for all $t \geq N_0$, where (78) follows from the degradability of Φ and the definition of $\eta(\mathcal{D})$, (79) follows from (5). The second to last inequality follows from $D(\rho\|\sigma) \geq \frac{1}{2}\mathrm{Tr}\left[(\rho-\sigma)^2\right]$ [21].

Denote $\frac{1-\eta(\mathcal{D})}{4\lambda_n}\,\mu_{min}$ by a, then $a>0$ and $D\left(\Phi\left(\rho^*\right)\|\Phi\left(\rho^{(t)}\right)\right)-D\left(\Phi_c\left(\rho^*\right)\|\Phi_c\left(\rho^{(t)}\right)\right) \geq a\,D\left(\rho^*\|\rho^{(t)}\right)$ for all $t \geq N_0$. Conbine this with (44) and (49), we have $I_{\mathrm{coh}}(\Phi)-I_{\mathrm{coh}}^{(t)}(\Phi) \leq \gamma\left(1-\frac{a}{\gamma}\right)^{t-N_0} D(\rho^*\|\rho^{(N_0)})$ holds for all $t \geq N_0$. □

4.2 Proof of Corollary 1

Case 1: Qubit dephasing channel

In this subsection we verify that the qubit dephasing channel Ψ_p satisfies the conditions in Theorem 1 in the required parameter regime.

Proof of injectivity of Ψ_p:

For $\rho = \begin{pmatrix}\rho_{00} & \rho_{01}\\ \rho_{10} & \rho_{11}\end{pmatrix}$, one has

$$\Psi_p(\rho) = \begin{pmatrix}\rho_{00} & (1-2p)\rho_{01}\\ (1-2p)\rho_{10} & \rho_{11}\end{pmatrix}. \tag{83}$$

Hence, if $p \neq \frac{1}{2}$, then $\Psi_p(\rho)=0$ implies $\rho_{00}=\rho_{11}=\rho_{01}=\rho_{10}=0$ and therefore $\rho=0$. Thus Ψ_p is injective for all $p \in \left(0,\frac{1}{2}\right)\cup\left(\frac{1}{2},1\right)$.

Proof of degradability of Ψ_p:

A Kraus representation of Ψ_p is given by

$$\Psi_p(\rho) = E_0^*\rho E_0 + E_1^*\rho E_1, \tag{84}$$

where

$$E_0 = \begin{pmatrix} \sqrt{1-p} & 0 \\ 0 & \sqrt{1-p} \end{pmatrix}, \quad E_1 = \begin{pmatrix} \sqrt{p} & 0 \\ 0 & -\sqrt{p} \end{pmatrix}. \tag{85}$$

Let $U = E_0 \otimes |0\rangle + E_1 \otimes |1\rangle$ be the Stinespring isometry associated with this Kraus representation [22]. Then the complementary channel is obtained by tracing out the output system, namely $\Psi_p^c(\rho) = \mathrm{Tr}_B\left(U\rho U^*\right)$. A direct computation gives

$$\Psi_p^c(\rho) = \begin{pmatrix} (1-p)(\rho_{00}+\rho_{11}) & \sqrt{p(1-p)}(\rho_{00}-\rho_{11}) \\ \sqrt{p(1-p)}(\rho_{00}-\rho_{11}) & p(\rho_{00}+\rho_{11}) \end{pmatrix}. \tag{86}$$

In particular, $\Psi_p^c(\rho)$ depends only on the diagonal entries of ρ. Since Ψ_p preserves the diagonal entries, we have $\Psi_p^c \circ \Psi_p = \Psi_p^c$. Thus the degrading channel may therefore be chosen as $\mathcal{D} = \Psi_p^c$ which implies Ψ_p is degradable for all $p \in (0,1)$.

Proof of the strict contraction property for the degrading channel $\mathcal{D}$ of Ψ_p:

By the preceding analysis, the complementary channel of Ψ_p can be chosen as its degrading channel. The proof of $\eta(\mathcal{D}) < 1$ is deferred to Appendix B.

Case 2: Qubit amplitude damping channel

We next verify the corresponding conditions for the qubit amplitude damping channel Φ_p.

Proof of injectivity of Φ_p:

For $\rho = \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix}$, one obtains

$$\Phi_p(\rho) = \begin{pmatrix} \rho_{00}+p\rho_{11} & \sqrt{1-p}\rho_{01} \\ \sqrt{1-p}\rho_{10} & (1-p)\rho_{11} \end{pmatrix}. \tag{87}$$

If $p \in [0,1)$, then $\Phi_p(\rho) = 0$ implies $\rho_{00} = \rho_{11} = \rho_{01} = \rho_{10} = 0$. Therefore $\rho = 0$, and Φ_p is injective for all $p \in [0,1)$.

Proof of degradability of Φ_p:

It is known that Φ_p is degradable for all $p \in [0, \frac{1}{2}]$ [23]. More precisely, under the standard identification of the output and environment systems, we have

$$\Phi_p^c = \Phi_{1-p} = \Phi_{\frac{1-2p}{1-p}} \circ \Phi_p. \tag{88}$$

Thus, for $p \in [0, \frac{1}{2}]$, a degrading channel of Φ_p is given by $\mathcal{D} = \Phi_{\frac{1-2p}{1-p}}$.

Proof of the strict contraction property for the degrading channel $\mathcal{D}$ of Φ_p:

For $p \in [0, \frac{1}{2})$, it follows from [18] that

$$\eta(\mathcal{D}) = \eta\left(\Phi_{\frac{1-2p}{1-p}}\right) \leq \sqrt{1 - \frac{1-2p}{1-p}} \tag{89}$$

$$= \sqrt{\frac{p}{1-p}} \tag{90}$$

$$< 1 \tag{91}$$

for all $p \in [0, \frac{1}{2})$.

5 Proof of Contraction coefficients for amplitude damping channels

5.1 Proof of Theorem 2

By Lemma 1, to prove Theorem 2 it suffices to establish the identity

$$\eta_{\mathrm{BKM}}(\Phi_p) := \sup_{\substack{\rho\in D(\mathcal{X})\\ X=X^*,\ \mathrm{Tr}(X)=0\\ X\neq 0}} \frac{g_{\Phi_p(\rho)}\left(\Phi_p(X)\right)}{g_\rho(X)} \tag{92}$$

$$= \sup_{r\in[0,1)} (1-p)\frac{r}{\operatorname{arctanh}(r)}\frac{\operatorname{arctanh}\left(p+(1-p)r\right)}{p+(1-p)r}. \tag{93}$$

First using the rotational invariance of amplitude damping channel around z-axis, we show that the maximization in (92) over all input states can be restricted to states whose Bloch vectors lie in the xoz-plane. Then using the special property of BKM metric induced by amplitude damping channel, we show that the maximization in (92) over all tangent vectors can be decomposed into two cases: one is the tangent vector is along the y-axis and the other case is that the tangent vector is on the xoz-plane. We then compute the the maximum value when the states are on the xoz-plane and the tangent vectors are along the y-axis and show that this maximum value dominate the other case.

Throughout this subsection, we write

$$\Theta(p) = \sup_{r\in[0,1)} (1-p)\frac{r}{\operatorname{arctanh}(r)}\frac{\operatorname{arctanh}\left(p+(1-p)r\right)}{p+(1-p)r}. \tag{94}$$

Using representations on Bloch ball, 93 is equivalent to $g_{\vec{w}_{\Phi_p}}(\vec{v}_{\Phi_p}) \leq \Theta(p)\, g_{\vec{w}}(\vec{v})$ for all $|\vec{w}| \leq 1$ and all $\vec{v} \in \mathbb{R}^3$. If $|\vec{w}| = 1$, namely if $\rho_{\vec{w}}$ is a pure state, then $g_{\vec{w}}(\vec{v}) = +\infty$ which implies the desired inequality is trivial. Therefore, throughout the rest of this subsection, we assume that $|\vec{w}| < 1$. Let

$$T_{\Phi_p} = \mathrm{diag}\left(\sqrt{1-p}, \sqrt{1-p}, 1-p\right), \quad t_{\Phi_p} = (0,0,p)\,. \tag{95}$$

Then in Bloch coordinates, the amplitude damping channel Φ_p maps $\vec{w} = (w_1, w_2, w_3)$ to $\vec{w}_{\Phi_p} = T_{\Phi_p}\vec{w} + t_{\Phi_p} = \left(\sqrt{1-p}\, w_1, \sqrt{1-p}\, w_2, p+(1-p)w_3\right)$ and maps $\vec{v} = (y_1, y_2, y_3)$ to $\vec{v}_{\Phi_p} = T_{\Phi_p}\vec{v} = \left(\sqrt{1-p}\, y_1, \sqrt{1-p}\, y_2, (1-p)y_3\right)$. For $p \in (0,1)$ and $|\vec{w}| < 1$, one also has $|\vec{w}_{\Phi_p}| < 1$ (The details are given in Appendix C).

The following lemma indicates that we can restrict the optimization range of w to the xoz-plane. That is, we just need to prove $g_{\vec{w}_{\Phi_p}}(\vec{v}_{\Phi_p}) \leq \Theta(p) g_{\vec{w}}(\vec{v})$ for all $\vec{v} \in \mathbb{R}^3$ and $\vec{w} \in xoz$-plane: $|\vec{w}| < 1$.

Lemma 3. *Let $p \in (0,1)$ and $C(p) \geq 0$. Suppose that $g_{\vec{w}_{\Phi_p}}(\vec{v}_{\Phi_p}) \leq C(p) g_{\vec{w}}(\vec{v})$ for all $\vec{v} \in \mathbb{R}^3$ and all $\vec{w} \in xoz - plane : |\vec{w}| < 1$, then $g_{\vec{w}_{\Phi_p}}(\vec{v}_{\Phi_p}) \leq C(p) g_{\vec{w}}(\vec{v})$ holds for all $\vec{v} \in \mathbb{R}^3$ and $\vec{w} \in \mathbb{R}^3 : |\vec{w}| < 1$.*

Proof. Let $\vec{w} = (w_1, w_2, w_3)$ be the Bloch vector of $\rho_{\vec{w}}$ and $\vec{v}$ be the Pauli-coordinate vector of $X_{\vec{v}}$. Then define

$$\theta = \begin{cases} \arctan\frac{w_2}{w_1}, & w_1 \neq 0, \\ \frac{\pi}{2}, & otherwise. \end{cases} \tag{96}$$

and

$$U = \begin{pmatrix} e^{-\frac{i\theta}{2}} & 0 \\ 0 & e^{\frac{i\theta}{2}} \end{pmatrix}. \tag{97}$$

It can be verified that U is a unitary matrix and $U\rho_{\vec{w}}U^* = \frac{1}{2}(I_2 + \vec{w}_U \cdot \vec{\sigma})$ where $\vec{w}_U = (cos\theta \cdot w_1, 0, w_3) \in xoz$-plane.

Using the unitary property of U we have that

$$(U\rho_{\vec{w}}U^* + rI)^{-1} = (U(\rho_{\vec{w}} + rI)U^*)^{-1} = U(\rho_{\vec{w}} + rI)^{-1}U^*, \tag{98}$$

which, together with the unitary invariance of the BKM metric, implies that

$$g_{\rho_{\vec{w}}}(X_{\vec{v}}) = g_{U\rho_{\vec{w}}U^*}(UX_{\vec{v}}U^*).$$

Notice that $\Phi_p(UXU^*) = U\Phi_p(X)U^*$ for all $X \in L(\mathcal{X})$. Then by direct calculations, we obtain

$$\begin{aligned}
g_{\vec{w}_{\Phi_p}}\left(\vec{v}_{\Phi_p}\right) &= g_{\Phi_p(\rho_{\vec{w}})}\left(\Phi_p(X_{\vec{v}})\right) && (99)\\
&= g_{U\Phi_p(\rho_{\vec{w}})U^*}\left(U\Phi_p(X_{\vec{v}})U^*\right) && (100)\\
&= g_{\Phi_p(U\rho_{\vec{w}}U^*)}\left(\Phi_p(UX_{\vec{v}}U^*)\right) && (101)\\
&\leq C(p)g_{U\rho_{\vec{w}}U^*}(UX_{\vec{v}}U^*) && (102)\\
&= C(p)g_{\rho_{\vec{w}}}(X_{\vec{v}}) && (103)\\
&= C(p)g_{\vec{w}}(\vec{v}). && (104)
\end{aligned}$$

for all $\vec{v} \in \mathbb{R}^3$ and $\vec{w} \in \mathbb{R}^3 : |\vec{w}| < 1$. □

Using formulas in Lemma 2, we have the follow decomposition:

$$\begin{aligned}
g_{\vec{w}}(\vec{v}) &= \frac{4|\vec{v}|^2}{1-|\vec{w}|^2}\left[\frac{|\vec{w}\cdot\vec{v}|^2}{|\vec{w}|^2|\vec{v}|^2}\left(1 - f(|\vec{w}|)\right) + f(|\vec{w}|)\right] && (105)\\
&= \frac{4}{1-|\vec{w}|^2}\left[\frac{1-f(|\vec{w}|)}{|\vec{w}|^2}(w_1y_1 + w_2y_2)^2 + \left(y_1^2 + y_2^2 + y_3^2\right)f(|\vec{w}|)\right] && (106)\\
&= g_{\vec{w}}(\vec{v}_y) + g_{\vec{w}}(\vec{v}_{xoz}) && (107)
\end{aligned}$$

where $\vec{v}_y = (0, y_2, 0) \in y$-axis and $\vec{v}_{xoz} \in xoz$-plane. This decomposition indicates that the problem can be decomposed into two parts: $\vec{v}$ on the $y - axis$ and y on the $xoz - plane$. These two cases are treated in Lemma 4 and Lemma 5, respectively.

Lemma 4. *For $\vec{w} \in xoz$-plane: $|\vec{w}| < 1$ and $\vec{v} \in y$-axis, one obtains $g_{\vec{w}_{\Phi_p}}\left(\vec{v}_{\Phi_p}\right) \leq \Theta(p)g_{\vec{w}}\left(\vec{v}\right)$.*

Proof. Let

$$\vec{w} = (w_1, 0, w_3) = (r\cos\theta, 0, r\sin\theta), \quad r \in [0, 1), \quad \theta \in [0, 2\pi), \tag{108}$$

and let

$$\vec{v} = (0, y_2, 0), \quad y_2 \in \mathbb{R}, \quad y_2 \neq 0. \tag{109}$$

A direct calculation gives $\vec{w} \cdot \vec{v} = 0$ and $\vec{w}_{\Phi_p} \cdot \vec{v}_{\Phi_p} = 0$. Then by Lemma 2, we have

$$l(p) \triangleq \sup_{\substack{\vec{w} \in xoz-plane: |\vec{w}|<1 \\ \vec{v} \in y-axis,\ \vec{v} \neq \vec{0}}} \frac{g_{\vec{w}_{\Phi_p}}\left(\vec{v}_{\Phi_p}\right)}{g_{\vec{w}}\left(\vec{v}\right)} \tag{110}$$

$$= \sup_{\substack{\vec{w} \in xoz-plane: |\vec{w}|<1 \\ \vec{v} \in y-axis,\ \vec{v} \neq \vec{0}}} \frac{\frac{4|\vec{v}_{\Phi_p}|^2}{1-|\vec{w}_{\Phi_p}|^2} f\left(|\vec{w}_{\Phi_p}|\right)}{\frac{4|\vec{v}|^2}{1-|\vec{w}|^2} f\left(|\vec{w}|\right)} \tag{111}$$

$$= \sup_{\vec{w} \in xoz-plane: |\vec{w}|<1} (1-p) \frac{1-|\vec{w}|^2}{1-|\vec{w}_{\Phi_p}|^2} \frac{f\left(|\vec{w}_{\Phi_p}|\right)}{f\left(|\vec{w}|\right)}. \tag{112}$$

Let $m_r(\theta) \triangleq |\vec{w}_{\Phi_p}|$. A direct computation gives

$$m_r(\theta) = \sqrt{(1-p) r^2 \cos^2\theta + \left(p + (1-p) r \sin\theta\right)^2} \tag{113}$$

$$= \sqrt{(1-p) r^2 \cos^2\theta + p^2 + 2p(1-p) r \sin\theta + (1-p)^2 r^2 \sin^2\theta}. \tag{114}$$

Using the expression of f in terms of artanh, we obtain

$$l(p) = \sup_{\substack{r \in [0,1) \\ \theta \in [0, 2\pi)}} (1-p) \frac{r}{\operatorname{artanh}(r)} \frac{\operatorname{artanh}\left(m_r(\theta)\right)}{m_r(\theta)}. \tag{115}$$

It remains to maximize the last factor with respect to θ. Since

$$m_r^2(\theta) = p^2 + (1-p) r^2 + 2p(1-p) r \sin\theta - p(1-p) r^2 \sin^2\theta, \tag{116}$$

$m_r^2(\theta)$ is a quadratic function of $\sin\theta$. For fixed $r \in [0,1)$, it attains its maximum on $[-1,1]$ at $\sin\theta = 1$. Hence

$$\max_{\theta \in [0, 2\pi)} m_r(\theta) = m_r\left(\frac{\pi}{2}\right) = p + (1-p) r. \tag{117}$$

Moreover, the function $x \longmapsto \frac{\operatorname{artanh}(x)}{x}$ is increasing on $[0,1)$. Therefore,

$$l(p) = \sup_{\substack{r \in [0,1) \\ \theta \in [0, 2\pi)}} (1-p) \frac{r}{\operatorname{artanh}(r)} \frac{\operatorname{artanh}\left(m_r(\theta)\right)}{m_r(\theta)}$$

$$= \sup_{r \in [0,1)} (1-p) \frac{r}{\operatorname{artanh}(r)} \frac{\operatorname{artanh}\left(p + (1-p) r\right)}{p + (1-p) r}$$

$$= \sup_{r \in [0,1)} (1-p) \frac{r}{(1-p) r + p} \frac{\operatorname{artanh}\left((1-p) r + p\right)}{\operatorname{artanh}(r)}. \tag{118}$$

This completes the proof. □

Lemma 5. *For $\vec{w} \in xoz\text{-plane}$: $|\vec{w}| < 1$ and $\vec{v} \in xoz\text{-plane}$, one obtains $g_{\vec{w}_{\Phi_p}}\left(\vec{v}_{\Phi_p}\right) \leq \Theta(p) g_{\vec{w}}\left(\vec{v}\right)$.*

Proof. As in the proof of Lemma 4, let

$$g(p) \triangleq \sup_{\substack{\vec{w}\in xoz-plane:|\vec{w}|<1 \\ \vec{v}\in xoz-plane,\ \vec{v}\neq\vec{0}}} \frac{g_{\vec{w}_{\Phi_p}}\left(\vec{v}_{\Phi_p}\right)}{g_{\vec{w}}\left(\vec{v}\right)} \tag{119}$$

$$= \sup_{\substack{\vec{w}\in xoz-plane:|\vec{w}|<1 \\ \vec{v}\in xoz-plane,\ \vec{v}\neq\vec{0}}} \frac{\vec{v}^T N \vec{v}}{\vec{v}^T M \vec{v}}. \tag{120}$$

Here

$$M = \frac{1}{1-|\vec{w}|^2}\left[f\left(|\vec{w}|\right) I + \frac{1-f\left(|\vec{w}|\right)}{|\vec{w}|^2}\vec{w}\vec{w}^T\right] \tag{121}$$

and

$$N = \frac{1}{1-|\vec{w}_{\Phi_p}|^2} T_{\Phi_p}^T \left[f\left(|\vec{w}_{\Phi_p}|\right) I + \frac{1-f\left(|\vec{w}_{\Phi_p}|\right)}{|\vec{w}_{\Phi_p}|^2}\vec{w}_{\Phi_p}\vec{w}_{\Phi_p}^T\right] T_{\Phi_p}. \tag{122}$$

More explicitly, since $\vec{w} = (w_1, 0, w_3)$, we have

$$M = \frac{1}{1-|\vec{w}|^2}\begin{pmatrix} \dfrac{f\left(|\vec{w}|\right) w_3^2 + w_1^2}{|\vec{w}|^2} & 0 & \dfrac{\left(1-f\left(|\vec{w}|\right)\right) w_1 w_3}{|\vec{w}|^2} \\ 0 & f\left(|\vec{w}|\right) & 0 \\ \dfrac{\left(1-f\left(|\vec{w}|\right)\right) w_1 w_3}{|\vec{w}|^2} & 0 & \dfrac{f\left(|\vec{w}|\right) w_1^2 + w_3^2}{|\vec{w}|^2} \end{pmatrix}. \tag{123}$$

Write $\vec{w}_{\Phi_p} = (\widetilde{w}_1, 0, \widetilde{w}_3)$ where $\widetilde{w}_1 = \sqrt{1-p}\, w_1$ and $\widetilde{w}_3 = p + (1-p) w_3$. Then we have $N = \frac{1}{1-|\vec{w}_{\Phi_p}|^2} N'$ where

$$N' = \begin{pmatrix} (1-p)\dfrac{f\left(|\vec{w}_{\Phi_p}|\right)\widetilde{w}_3^2 + \widetilde{w}_1^2}{|\vec{w}_{\Phi_p}|^2} & 0 & (1-p)^{\frac{3}{2}}\dfrac{\left(1-f\left(|\vec{w}_{\Phi_p}|\right)\right)\widetilde{w}_1\widetilde{w}_3}{|\vec{w}_{\Phi_p}|^2} \\ 0 & (1-p) f\left(|\vec{w}_{\Phi_p}|\right) & 0 \\ (1-p)^{\frac{3}{2}}\dfrac{\left(1-f\left(|\vec{w}_{\Phi_p}|\right)\right)\widetilde{w}_1\widetilde{w}_3}{|\vec{w}_{\Phi_p}|^2} & 0 & (1-p)^2\dfrac{f\left(|\vec{w}_{\Phi_p}|\right)\widetilde{w}_1^2 + \widetilde{w}_3^2}{|\vec{w}_{\Phi_p}|^2} \end{pmatrix}. \tag{124}$$

For each fixed $\vec{w}$, it is obvious that

$$\sup_{\substack{\vec{v}\in xoz-plane \\ \vec{v}\neq\vec{0}}} \frac{\vec{v}^T N \vec{v}}{\vec{v}^T M \vec{v}} \leq \lambda_{\max}\left(M^{-\frac{1}{2}} N M^{-\frac{1}{2}}\right) = \lambda_{\max}\left(M^{-1}N\right). \tag{125}$$

The eigenvalues of $M^{-1}N$ can be characterized by

$$\lambda \in \sigma\left(M^{-1}N\right) \quad \Longleftrightarrow \quad \det\left(\lambda M - N\right) = 0. \tag{126}$$

From the block structure of M and N, the eigenvalues of $M^{-1}N$ consist of one eigenvalue λ_y corresponding to the y-axis and two eigenvalues $\lambda_\pm$ corresponding to the xoz-plane. A direct calculation gives

$$\lambda_{\max}\left(M^{-1}N\right) = \lambda_y = (1-p)\,\frac{1-|\vec{w}|^2}{1-|\vec{w}_{\Phi_p}|^2}\,\frac{f\left(|\vec{w}_{\Phi_p}|\right)}{f\left(|\vec{w}|\right)}. \tag{127}$$

The details of this spectral comparison are deferred to Appendix D. Consequently, by Lemma 4, we have

$$\begin{aligned} g(p) &\leq \sup_{\vec{w}\in xoz-plane:|\vec{w}|<1} \lambda_{\max}\left(M^{-1}N\right) && (128)\\ &\leq \sup_{\vec{w}\in xoz-plane:|\vec{w}|<1} (1-p)\,\frac{1-|\vec{w}|^2}{1-|\vec{w}_{\Phi_p}|^2}\,\frac{f\left(|\vec{w}_{\Phi_p}|\right)}{f\left(|\vec{w}|\right)} \\ &\leq \sup_{r\in(0,1)} (1-p)\,\frac{r}{(1-p)\,r+p}\,\frac{\operatorname{artanh}\left((1-p)\,r+p\right)}{\operatorname{artanh}\left(r\right)}. && (129) \end{aligned}$$

This completes the proof. □

Combining the above two lemmas, we have

$$\begin{aligned} \sup_{\substack{\vec{w}\in xoz-plane:|\vec{w}|<1\\ \vec{v}\in\mathbb{R}^3:\vec{v}\neq\vec{0}}} \frac{g_{\vec{w}_{\Phi_p}}(\vec{v}_{\Phi_p})}{g_{\vec{w}}(\vec{v})} &= \sup_{\substack{\vec{w}\in xoz-plane:|\vec{w}|<1\\ \vec{v}\in\mathbb{R}^3:\vec{v}\neq\vec{0}}} \frac{g_{\vec{w}_{\Phi_p}}\left(\left(\vec{v}_{\Phi_p}\right)_y\right) + g_{\vec{w}_{\Phi_p}}\left(\left(\vec{v}_{\Phi_p}\right)_{xoz}\right)}{g_{\vec{w}}(\vec{v}_y) + g_{\vec{w}}(\vec{v}_{xoz})} && (130)\\ &\leq \Theta(p). && (131) \end{aligned}$$

5.2 Proof of Theorem 3

For $p_2 > p_1$, the channel Φ_{p_2} can be decomposed as

$$\Phi_{p_2} = \Phi_q \circ \Phi_{p_1}, \quad q = \frac{p_2 - p_1}{1-p_1}. \tag{132}$$

Therefore, the comparison between Φ_{p_1} and Φ_{p_2} is equivalent to studying the contraction induced by the additional amplitude damping channel Φ_q on the image of Φ_{p_1} instead of the whole Bloch ball.

By Lemma 1, to prove Theorem 3 it suffices to establish the identity

$$\begin{aligned} \eta_{\mathrm{BKM}}(\Phi_{p_2}, \Phi_{p_1}) &:= \sup_{\substack{\rho\in D(\mathcal{X})\\ X=X^*,\ \mathrm{Tr}(X)=0\\ X\neq 0}} \frac{g_{\Phi_{p_2}(\rho)}\left(\Phi_{p_2}(X)\right)}{g_{\Phi_{p_1}(\rho)}\left(\Phi_{p_1}(X)\right)} && (133)\\ &= \sup_{r\in I_{p_1}} (1-q)\frac{r}{\operatorname{arctanh}(r)}\,\frac{\operatorname{arctanh}\left(q+(1-q)r\right)}{q+(1-q)r} && (134) \end{aligned}$$

Let $\omega = \Phi_{p_1}(\rho)$ and $Y = \Phi_{p_1}(X)$. Since $p_1 \in (0,1)$, the linear map induced by Φ_{p_1} on traceless Hermitian operators is invertible. Therefore, as X ranges over all traceless Hermitian operators, Y also ranges over all traceless Hermitian operators. Thus the problem is equivalent to estimating

$$g_{\Phi_q(\omega)}(\Phi_q(Y)) \leq c \cdot g_\omega(Y) \tag{135}$$

for all $\omega \in \Phi_{p_1}\left(D(\mathcal{X})\right)$ and all traceless Hermitian operators Y.

We now pass to Bloch coordinates. Write the Bloch vector of ω as $\vec{w}$ and the Pauli-coordinate vector of Y as $\vec{v}$. The image $\Phi_{p_1}\left(D(\mathcal{X})\right)$ is the ellipsoid

$$A_{p_1} = \left\{ \vec{w}(w_1, w_2, w_3) \in \mathbb{R}^3 : \frac{w_1^2 + w_2^2}{1-p_1} + \frac{(w_3 - p_1)^2}{(1-p_1)^2} \leq 1 \right\}. \tag{136}$$

If $|\vec{w}| = 1$, then ω is a pure state, and the corresponding BKM metric gives a trivial inequality. Hence, in the following, we assume that $|\vec{w}| < 1$.

By the rotational invariance around the z-axis, as in Lemma 5, it suffices to restrict the Bloch vector $\vec{w}$ to the xoz-plane. Therefore, we only need to consider $\vec{w} = (w_1, 0, w_3) \in A_{p_1}$ where this condition is equivalent to $\frac{w_1^2}{1-p_1} + \frac{(w_3-p_1)^2}{(1-p_1)^2} \leq 1$. Consequently, the desired relative contraction coefficient is reduced to

$$\Pi(q) = \sup_{\substack{\vec{w}\in A \\ \vec{v}\in\mathbb{R}^3 : \vec{v}\neq 0}} \frac{g_{\vec{w}_{\Phi_q}}\left(\vec{v}_{\Phi_q}\right)}{g_{\vec{w}}(\vec{v})}. \tag{137}$$

where $A = \left\{ \vec{w} = (w_1, 0, w_3) \in \mathbb{R}^3 : \frac{w_1^2}{1-p_1} + \frac{(w_3-p_1)^2}{(1-p_1)^2} \leq 1 \right\}$

The same with proof of Theorem 2, we can decompose $g_{\vec{w}}(\vec{v})$ and $g_{\vec{w}_{\Phi_q}}(\vec{v}_{\Phi_q})$ into two parts respectively. When $\vec{v}$ on the y-axis, we derive that

$$\sup_{\substack{\vec{w}\in A \\ \vec{v}\in y-axis : \vec{v}\neq\ 0}} \frac{g_{\vec{w}_{\Phi_q}}(\vec{v}_{\Phi_q})}{g_{\vec{w}}(\vec{v})} = \sup_{\vec{w}\in A}(1-q)\frac{1-|\vec{w}|^2}{1-|\vec{w}_{\Phi_q}|^2}\frac{f(|\vec{w}_{\Phi_q}|)}{f(|\vec{w}|)}. \tag{138}$$

While when $\vec{v}$ on the xoz-plane, we have

$$\sup_{\substack{\vec{w}\in A \\ \vec{v}\in xoz-plane}} \frac{g_{\vec{w}_{\Phi_q}}(\vec{v}_{\Phi_q})}{g_{\vec{w}}(\vec{v})} \leq \sup_{\vec{w}\in A} \lambda_{max}(M^{-1}N) \tag{139}$$

$$\leq \sup_{\vec{w}\in A}(1-q)\frac{1-|\vec{w}|^2}{1-|\vec{w}_{\Phi_q}|^2}\frac{f(|\vec{w}_{\Phi_q}|)}{f(|\vec{w}|)} \tag{140}$$

where M and N are matrices defined in 5.1. In conclusion,

$$\Pi(q) = \sup_{\vec{w}\in A}(1-q)\frac{1-|\vec{w}|^2}{1-|\vec{w}_{\Phi_q}|^2}\frac{f\left(|\vec{w}_{\Phi_q}|\right)}{f\left(|\vec{w}|\right)} \tag{141}$$

$$= \sup_{\vec{w}\in A}(1-q)\frac{\operatorname{artanh}\left(|\vec{w}_{\Phi_q}|\right)}{|\vec{w}_{\Phi_q}|}\frac{|\vec{w}|}{\operatorname{artanh}\left(|\vec{w}|\right)} \tag{142}$$

Let $\vec{w} = (w_1, 0, w_3) = (r \cdot \cos\theta, 0, r \cdot \sin\theta)$ and hence $\vec{w}_{\Phi_q} = (\sqrt{1-q}\, w_1, 0, q + (1-q)w_3) = (\sqrt{1-q}\, r \cdot \cos\theta, 0, q + (1-q)r \cdot \sin\theta)$. Define $m_r(\theta) = |\vec{w}_{\Phi_q}|$. Then

$$\Pi(q) = \sup_{r,\theta}(1-q)\frac{\operatorname{artanh}\left(m_r(\theta)\right)}{m_r(\theta)}\frac{r}{\operatorname{artanh}(r)} \tag{143}$$

$$= \sup_{r}(1-q)\frac{r}{\operatorname{artanh}(r)}\sup_{\theta}\frac{\operatorname{artanh}\left(m_r(\theta)\right)}{m_r(\theta)} \tag{144}$$

It can be calculated that

$$m_r^2(\theta) = q^2 + 2q(1-q)r\sin\theta + (1-q)r^2\left(1 - q\sin^2\theta\right). \tag{145}$$

Viewing $m_r^2(\theta)$ as a function of $\sin\theta$, we have $\sup_{\theta} m_r(\theta) = m_r(\frac{\pi}{2}) = q + (1-q)r$. Combining it with the fact that $\frac{\operatorname{artanh}(x)}{x}$ is monotone increasing with respect to $x \in [0,1)$, we obtain the following result:

$$\Pi(q) = \sup_{r}(1-q)\frac{r}{\operatorname{artanh}(r)}\frac{\operatorname{artanh}\left(q + (1-q)r\right)}{q + (1-q)r} \tag{146}$$

where the supremum is taken over $[0,1)$ for $p_1 \in (0, \frac{1}{2}]$ and $[2p_1 - 1, 1)$ for $p_1 \in (\frac{1}{2}, 1)$.

5.3 Proof of Theorem 4

By Lemma 1, it suffices to prove

$$\check{\eta}_{\mathrm{BKM}}\left(\Phi_{p_2}, \Phi_{p_1}\right) := \inf_{\substack{\rho\in D(\mathcal{X}) \\ X=X^*,\ \mathrm{Tr}(X)=0 \\ X\neq 0}} \frac{g_{\Phi_{p_2}(\rho)}\left(\Phi_{p_2}(X)\right)}{g_{\Phi_{p_1}(\rho)}\left(\Phi_{p_1}(X)\right)} \tag{147}$$

$$= \frac{p_1(1-p_2)}{p_2(1-p_1)}. \tag{148}$$

The same with proof of Theorem 3, the above optimization is equivalent to

$$\check{\eta}_{\mathrm{BKM}}\left(\Phi_{p_2}, \Phi_{p_1}\right) = \inf_{\substack{\omega\in\Phi_{p_1}(D(\mathcal{X})) \\ Y=Y^*,\ \mathrm{Tr}(Y)=0 \\ Y\neq 0}} \frac{g_{\Phi_q(\omega)}\left(\Phi_q(Y)\right)}{g_{\omega}(Y)}. \tag{149}$$

and it suffices to restrict the Bloch vector $\vec{w}$ to the xoz-plane. Therefore, we only need to consider $\vec{w} = (w_1, 0, w_3) \in A_{p_1}$. Consequently, the relative expansion problem is reduced to computing

$$\Omega(q) = \inf_{\substack{\vec{w}\in A \\ \vec{v}\neq 0}} \frac{g_{\vec{w}_{\Phi_q}}\left(\vec{v}_{\Phi_q}\right)}{g_{\vec{w}}(\vec{v})}. \tag{150}$$

We also decompose $g_{\vec{w}}(\vec{v})$ and $g_{\vec{w}_{\Phi_q}}(\vec{v}_{\Phi_q})$ into two parts respectively. When $\vec{v}$ on the y-axis, we obtain

$$\inf_{\substack{\vec{w}\in A\\ \vec{v}\in y-axis:\vec{v}\neq\vec{0}}} \frac{g_{\vec{w}_{\Phi_q}}(\vec{v}_{\Phi_q})}{g_{\vec{w}}(\vec{v})} = \inf_{\vec{w}\in A}(1-q)\frac{1-|\vec{w}|^2}{1-|\vec{w}_{\Phi_q}|^2}\frac{f\left(|\vec{w}_{\Phi_q}|\right)}{f\left(|\vec{w}|\right)}. \tag{151}$$

By proof of Theorem 3, the eigenvalues corresponding to the restriction of $M^{-1}N$ to the xoz-plane are $\lambda_+\left(M^{-1}N\right)$ and $\lambda_-\left(M^{-1}N\right)$. In particular, the eigenvector associated with $\lambda_-\left(M^{-1}N\right)$ lies in the xoz-plane. Thus, for each fixed $\vec{w}\in A$, we have

$$\inf_{\substack{\vec{w}\in A\\ \vec{v}\in xoz-plane:\vec{v}\neq\vec{0}}} \frac{g_{\vec{w}_{\Phi_q}}(\vec{v}_{\Phi_q})}{g_{\vec{w}}(\vec{v})} = \inf_{\vec{w}\in A}\lambda_-(M^{-1}N). \tag{152}$$

In conclusion,

$$\Omega(q) = \inf_{\vec{w}\in A}\lambda_-(M^{-1}N) \tag{153}$$

$$= \inf_{\vec{w}\in A}\frac{1-|\vec{w}|^2}{1-|\vec{w}_{\Phi_q}|^2}\frac{C-\sqrt{C^2-4(1-p)^3 f\left(|\vec{w}|\right) f\left(|\vec{w}_{\Phi_q}|\right)}}{2f\left(|\vec{w}|\right)} \tag{154}$$

where

$$C = \frac{f(|\vec{w}|)w_3^2+w_1^2}{|\vec{w}|^2}(1-q)^2\frac{f(|\vec{w}_{\Phi_q}|)\widetilde{w}_1^2+\widetilde{w}_3^2}{|\vec{w}_{\Phi_q}|^2} \tag{155}$$

$$+\frac{f(|\vec{w}|)w_1^2+w_3^2}{|\vec{w}|^2}(1-q)\frac{f(|\vec{w}_{\Phi_q}|)\widetilde{w}_3^2+\widetilde{w}_1^2}{|\vec{w}_{\Phi_q}|^2} \tag{156}$$

$$-2\frac{\big(1-f(|\vec{w}|)\big)w_1w_3}{|\vec{w}|^2}(1-q)^{\frac{3}{2}}\frac{\big(1-f(|\vec{w}_{\Phi_q}|)\big)\widetilde{w}_1\widetilde{w}_3}{|\vec{w}_{\Phi_q}|^2}. \tag{157}$$

It can be calculated that

$$\inf_{\vec{w}\in A}\lambda_-(M^{-1}N) = \frac{p_1(1-p_2)}{p_2(1-p_1)} \tag{158}$$

whose proof is deferred to appendix E.

A Kraus Representation

Let $\mathcal{X}$ and $\mathcal{Y}$ be finite-dimensional Hilbert spaces. Then $\Phi\in T(\mathcal{X},\mathcal{Y})$ is a quantum channel if and only if there exist an alphabet Σ and a collection $\{A_a : a\in\Sigma\}\subseteq L(\mathcal{X},\mathcal{Y})$ satisfying:

$$\sum_{a\in\Sigma}A_a^*A_a = id_{\mathcal{X}} \quad \text{and} \quad \Phi(X) = \sum_{a\in\Sigma}A_a^*XA_a \tag{159}$$

for all $X\in L(\mathcal{X})$. The expression is called Kraus representation of Φ.

Next theorem demonstrates the relationship between the injectivity of Φ and its Kraus representation.

Theorem 5. *A quantum channel Φ is injective if and only if $\sum_{a\in\Sigma} A_a \otimes \overline{A_a}$ is full column rank.*

Proof. Notice that

$$\begin{aligned}
\mathrm{vec}\left(\Phi(X)\right) &= \mathrm{vec}(\sum_{a\in\Sigma} A_a^* X A_a) && (160)\\
&= \sum_{a\in\Sigma} \mathrm{vec}(A_a^* X A_a) && (161)\\
&= \sum_{a\in\Sigma} (A_a \otimes \overline{A_a})\mathrm{vec}(X) && (162)\\
&= (\sum_{a\in\Sigma} A_a \otimes \overline{A_a})\mathrm{vec}(X). && (163)
\end{aligned}$$

Denote $\sum_{a\in\Sigma} A_a \otimes \overline{A_a}$ by M. Then we have

$$\begin{aligned}
\Phi \text{ is injective} &\Leftrightarrow \Phi(X) = 0 \Rightarrow X = 0, \quad \text{for all } X \in L(\mathcal{X}) && (164)\\
&\Leftrightarrow \mathrm{vec}\left(\Phi(X)\right) = 0 \Rightarrow \mathrm{vec}(X) = 0, \quad \text{for all } X \in L(\mathcal{X}) && (165)\\
&\Leftrightarrow Mu = 0 \Rightarrow u = 0, \quad \text{for all } u \in \mathcal{X}\otimes\mathcal{X} && (166)\\
&\Leftrightarrow M \text{ is of full column rank.} && (167)
\end{aligned}$$

□

B Strict contraction of the degrading channel for the qubit dephasing channel

Let $\mathcal{X} = \mathbb{C}^{\{0,1\}}$. The completely dephasing channel is defined by

$$\Delta(X) = X(0,0)E_{0,0} + X(1,1)E_{1,1}, \quad X \in L(\mathcal{X}). \tag{168}$$

Hence,

$$\begin{aligned}
\eta(\mathcal{D}) &= \sup_{\substack{\rho\neq\sigma\\ \rho\ll\sigma}} \frac{D\left(\mathcal{D}(\rho)\|\mathcal{D}(\sigma)\right)}{D(\rho\|\sigma)} && (169)\\
&= \sup_{\substack{\rho\neq\sigma\\ \rho\ll\sigma}} \frac{D\left(\mathcal{D}\left(\Delta(\rho)\right)\|\mathcal{D}\left(\Delta(\sigma)\right)\right)}{D(\rho\|\sigma)} && (170)\\
&\leq \sup_{\substack{\rho\neq\sigma\\ \rho\ll\sigma}} \frac{D\left(\mathcal{D}\left(\Delta(\rho)\right)\|\mathcal{D}\left(\Delta(\sigma)\right)\right)}{D\left(\Delta(\rho)\|\Delta(\sigma)\right)} && (171)\\
&\leq \sup_{\substack{\rho\neq\sigma,\ \rho\ll\sigma\\ \rho,\sigma \text{ are diagonal}}} \frac{D\left(\mathcal{D}(\rho)\|\mathcal{D}(\sigma)\right)}{D(\rho\|\sigma)}. && (172)
\end{aligned}$$

Write

$$\rho = \begin{pmatrix} a & 0 \\ 0 & 1-a \end{pmatrix}, \quad \sigma = \begin{pmatrix} b & 0 \\ 0 & 1-b \end{pmatrix}, \quad a, b \in [0,1]. \tag{173}$$

For such diagonal states, the condition $\rho \ll \sigma$ is equivalent to $\mathrm{supp}(\rho) \subseteq \mathrm{supp}(\sigma)$. In particular, if $a \in (0,1)$, then ρ has full support. Hence $\rho \ll \sigma$ holds if and only if σ also has full support, that is, $b \in (0,1)$.

Define two states $\rho_0 = \begin{pmatrix} 1-p & -\sqrt{p(1-p)} \\ -\sqrt{p(1-p)} & p \end{pmatrix}$ and $\rho_1 = \begin{pmatrix} 1-p & \sqrt{p(1-p)} \\ \sqrt{p(1-p)} & p \end{pmatrix}$. Then $\mathcal{D}(\rho) = (1-a)\rho_0 + a\rho_1$. For simplicity, write

$$M(a) := \mathcal{D}(\rho) = \begin{pmatrix} 1-p & (2a-1)\sqrt{p(1-p)} \\ (2a-1)\sqrt{p(1-p)} & p \end{pmatrix} \tag{174}$$

and similarly write $M(b) := \mathcal{D}(\sigma)$.

Use the notation $D(a\|b) := a\log\frac{a}{b} + (1-a)\log\frac{1-a}{1-b}$. Then the ratio $\frac{D(M(a)\|M(b))}{D(a\|b)}$ is continuous on the set $\Omega := \{(a,b) \in (0,1)^2 : a \neq b\}$. Therefore, it remains to prove that

$$\sup_{(a,b)\in\Omega} \frac{D\left(M(a)\|M(b)\right)}{D(a\|b)} < 1. \tag{175}$$

Define $F(a) = \mathrm{Tr}\left[M(a)\log M(a)\right]$ and $G(a) = a\log a + (1-a)\log(1-a)$. Then it is easy to prove that $D(a\|b) = G(a) - G(b) - G'(b)(a-b)$. We then prove that $D\left(M(a)\|M(b)\right) = F(a) - F(b) - F'(b)(a-b)$, and it is obvious that we just need to show $F'(a) = \mathrm{Tr}\left[(\rho_1 - \rho_0)\log M(a)\right]$.

Suppose $M(a)$ has the spectral decomposition $M(a) = \sum_{i=1}^{2} \lambda_i(a) x_i(a) x_i(a)^*$ then it can be calculated that $\ln\left[2 \cdot F(a)\right] = \sum_{i=1}^{2} \lambda_i(a)\ln\lambda_i(a)$ and $\ln\left[2 \cdot F'(a)\right] = \sum_{i=1}^{2} \lambda_i'(a)\left(\ln\lambda_i(a) + 1\right)$. From the characteristic equation: $M(a)x_i(a) = \lambda_i(a)x_i(a)$, we can derive that $M'(a)x_i(a) + M(a)x_i'(a) = \lambda_i'(a)x_i(a) + \lambda_i(a)x_i'(a)$. A direct computation gives

$$\begin{aligned}
&\langle x_i(a), M'(a)x_i(a)\rangle + \langle x_i(a), M(a)x_i'(a)\rangle && (176)\\
&= \langle x_i(a), M'(a)x_i(a)\rangle + \langle M(a)x_i(a), x_i'(a)\rangle && (177)\\
&= \langle x_i(a), M'(a)x_i(a)\rangle + \lambda_i(a)\langle x_i(a), x_i'(a)\rangle && (178)\\
&= \langle x_i(a), \lambda_i'(a)x_i(a)\rangle + \lambda_i(a)\langle x_i(a), x_i'(a)\rangle && (179)\\
&= \lambda_i'(a) && (180)
\end{aligned}$$

which implies that $\lambda_i'(a) = \langle x_i(a), M'(a)x_i(a)\rangle$.

It follows that

$$\ln\left\{2\cdot\operatorname{Tr}\left[M'(a)\ln M(a)\right]\right\}=\ln\left\{2\cdot\operatorname{Tr}\left[M'(a)\left(\ln M(a)+I\right)\right]\right\} \tag{181}$$

$$=\sum_{i=1}^{2}\left(\ln\lambda_i(a)+1\right)\left\langle x_i(a),M'(a)x_i(a)\right\rangle \tag{182}$$

$$=\sum_{i=1}^{2}\left(\ln\lambda_i(a)+1\right)\lambda_i'(a) \tag{183}$$

$$=\ln\left[2\cdot F'(a)\right] \tag{184}$$

which implies that $F'(a)=\operatorname{Tr}\left[M'(a)\ln M(a)\right]=\operatorname{Tr}\left[(\rho_1-\rho_0)\log M(a)\right]$.

For any C^2 function H, one has

$$H(a)-H(b)-H'(b)(a-b)=\int_b^a\left[H'(t)-H'(b)\right]dt \tag{185}$$

$$=\int_b^a\int_b^t H''(s)dsdt. \tag{186}$$

Hence, we can infer that $D(M(a)\|M(b))\leq C\cdot D(a\|b)$ from $F''(t)\leq C\cdot G''(t)$ for $t\in(0,1)$ where C is a constant. Hence, the desired claim reduces to proving $\sup\limits_{t\in(0,1)}\frac{F''(t)}{G''(t)}<1$.

Let

$$\theta:=2t-1, \tag{187}$$

$$z:=|1-2p|, \tag{188}$$

$$k:=4p(1-p)=1-z^2, \tag{189}$$

$$u:=\sqrt{z^2+k\theta^2}. \tag{190}$$

By direct calculation, we obtain the eigenvalues of $M(a)$ are $\lambda_{1,2}(a)=\frac{1\pm u}{2}$. Hence

$$\ln\left[2\cdot F(a)\right]=\frac{1+u}{2}\ln\frac{1+u}{2}+\frac{1-u}{2}\ln\frac{1-u}{2} \tag{191}$$

$$=\frac{1}{2}\ln\frac{1-u^2}{4}+u\cdot\operatorname{artanh}(u). \tag{192}$$

A straightforward calculation yields

$$\frac{F''(t)}{G''(t)}=k\left[\frac{z^2(1-t^2)}{u^3}\operatorname{artanh}(u)+\frac{t^2}{u^2}\right]. \tag{193}$$

Write

$$\Gamma(u)=k\left[\frac{z^2(1-t^2)}{u^3}\operatorname{artanh}(u)+\frac{t^2}{u^2}\right] \tag{194}$$

$$=1-\frac{z^2}{u^2}+\frac{z^2(1-u^2)}{u^3}\operatorname{artanh}(u) \tag{195}$$

as a function of $u \in [z, 1)$. Next we prove that Γ is decreasing on $(0, 1)$ and hence

$$\Gamma(u) \leq \Gamma(z) \tag{196}$$

$$= 4p(1-p)\frac{\operatorname{artanh}(|1-2p|)}{|1-2p|} \tag{197}$$

$$< 1. \tag{198}$$

A direct differentiation gives $\Gamma'(u) = \frac{z^2}{u^4}\left(u^2 \operatorname{artanh}(u) + 3u - 3\operatorname{artanh}(u)\right)$. Thus, it remains to determine the sign of $h(u) := u^2 \operatorname{artanh}(u) + 3u - 3\operatorname{artanh}(u)$. Differentiating h, we obtain

$$h'(u) = 2u \operatorname{artanh}(u) + \frac{u^2}{1-u^2} + 3 - \frac{3}{1-u^2} \tag{199}$$

$$= 2u \operatorname{artanh}(u) - \frac{2u^2}{1-u^2} \tag{200}$$

$$= 2u\left(\operatorname{artanh}(u) - \frac{u}{1-u^2}\right). \tag{201}$$

For $0 < u < 1$, we have

$$\operatorname{artanh}(u) = \int_0^u \frac{dt}{1-t^2} < \int_0^u \frac{dt}{1-u^2} = \frac{u}{1-u^2}. \tag{202}$$

It follows that $h'(u) < 0$ for all $u \in (0, 1)$. Since $h(0) = 0$, we conclude that $\Gamma'(u) < 0$ for all $u \in (0, 1)$. Consequently, we obtain

$$\Gamma'(u) < 0, \quad 0 < u < 1. \tag{203}$$

C A useful property of the amplitude damping channel

In this appendix, we prove that for any $p \in (0, 1)$ and any $\vec{w} \in \mathbb{R}^3$ with $|\vec{w}| < 1$, the transformed Bloch vector $\vec{w}_{\Phi_p}$ also satisfies $|\vec{w}_{\Phi_p}| < 1$.

Recall that the qubit amplitude damping channel Φ_p acts on the Bloch vector $\vec{w} = (w_1, w_2, w_3)$ as $\vec{w}_{\Phi_p} = \left(\sqrt{1-p}\, w_1, \sqrt{1-p}\, w_2, p + (1-p)w_3\right)$. Therefore,

$$|\vec{w}_{\Phi_p}|^2 = (1-p)w_1^2 + (1-p)w_2^2 + (p + (1-p)w_3)^2 \tag{204}$$

$$= (1-p)(w_1^2 + w_2^2) + p^2 + 2p(1-p)w_3 + (1-p)^2 w_3^2. \tag{205}$$

Hence,

$$1 - |\vec{w}_{\Phi_p}|^2 = 1 - (1-p)(w_1^2 + w_2^2) - p^2 - 2p(1-p)w_3 - (1-p)^2 w_3^2 \tag{206}$$

$$= (1-p)\left[1 + p - (w_1^2 + w_2^2) - 2pw_3 - (1-p)w_3^2\right] \tag{207}$$

$$= (1-p)\left[1 - |\vec{w}|^2 + p(1-w_3)^2\right]. \tag{208}$$

Since $p \in (0, 1)$, we have $1 - p > 0$. Moreover, $|\vec{w}| < 1$ implies $1 - |\vec{w}|^2 > 0$ and clearly $p(1-w_3)^2 \geq 0$. It follows that

$$1 - |\vec{w}_{\Phi_p}|^2 = (1-p)\left[1 - |\vec{w}|^2 + p(1-w_3)^2\right] > 0. \tag{209}$$

Consequently, we derive that $|\vec{w}_{\Phi_p}| < 1$.

D Details of the spectral comparison

In this appendix we prove the spectral comparison used in the proof of the contraction coefficient for the amplitude damping channel. Throughout this appendix, let

$$\eta = 1 - p. \tag{210}$$

For $\vec{w} = (w_1, 0, w_3)$, write

$$r = |\vec{w}|, \qquad \vec{w}_{\Phi_p} = (\widetilde{w}_1, 0, \widetilde{w}_3)\,, \tag{211}$$

where

$$\widetilde{w}_1 = \sqrt{\eta} w_1, \qquad \widetilde{w}_3 = p + \eta w_3. \tag{212}$$

We also write

$$r_{\Phi_p} = |\vec{w}_{\Phi_p}|. \tag{213}$$

All formulas below are understood by continuity at $r = 0$ or $r_{\Phi_p} = 0$.

Recall that

$$f(x) = \left(1 - x^2\right) \frac{\operatorname{arctanh} x}{x}, \qquad 0 \le x < 1, \tag{214}$$

with the continuous extension $f(0) = 1$. For simplicity, put

$$f_r = f(r), \qquad f_\Phi = f\left(r_{\Phi_p}\right). \tag{215}$$

From the block structure of M and N, the y-direction decouples from the xoz-plane. The generalized eigenvalue corresponding to the y-direction is

$$\lambda_y = \eta \frac{1 - r^2}{1 - r_{\Phi_p}^2} \frac{f_\Phi}{f_r}. \tag{216}$$

It remains to show that the two generalized eigenvalues coming from the xoz-plane are bounded above by λ_y.

Let

$$B = \begin{pmatrix} \dfrac{f_r w_3^2 + w_1^2}{r^2} & \dfrac{(1 - f_r)\, w_1 w_3}{r^2} \\ \dfrac{(1 - f_r)\, w_1 w_3}{r^2} & \dfrac{f_r w_1^2 + w_3^2}{r^2} \end{pmatrix} \tag{217}$$

and

$$C = \begin{pmatrix} \eta \dfrac{f_\Phi \widetilde{w}_3^2 + \widetilde{w}_1^2}{r_{\Phi_p}^2} & \eta^{\frac{3}{2}} \dfrac{(1 - f_\Phi)\, \widetilde{w}_1 \widetilde{w}_3}{r_{\Phi_p}^2} \\ \eta^{\frac{3}{2}} \dfrac{(1 - f_\Phi)\, \widetilde{w}_1 \widetilde{w}_3}{r_{\Phi_p}^2} & \eta^2 \dfrac{f_\Phi \widetilde{w}_1^2 + \widetilde{w}_3^2}{r_{\Phi_p}^2} \end{pmatrix}. \tag{218}$$

Then the xoz-blocks of M and N are respectively

$$\frac{1}{1-r^2}B, \qquad \frac{1}{1-r_{\Phi_p}^2}C. \tag{219}$$

Therefore the two generalized eigenvalues corresponding to the xoz-plane are

$$\lambda_\pm = \frac{1-r^2}{1-r_{\Phi_p}^2}\mu_\pm, \tag{220}$$

where $\mu_\pm$ are the two eigenvalues of

$$A = B^{-1}C. \tag{221}$$

Thus it suffices to prove

$$\mu_\pm \le \mu_y, \qquad \mu_y = \eta\frac{f_\Phi}{f_r}. \tag{222}$$

We first compute the determinant. Since B has eigenvalues 1 and f_r, one has

$$\det B = f_r. \tag{223}$$

Similarly, since C is obtained from the matrix

$$f_\Phi I + \frac{1-f_\Phi}{r_{\Phi_p}^2}\vec{w}_{\Phi_p}\vec{w}_{\Phi_p}^T \tag{224}$$

by the congruence with $\mathrm{diag}\left(\sqrt{\eta},\eta\right)$ on the xoz-plane, we have

$$\det C = \eta^3 f_\Phi. \tag{225}$$

Hence

$$\det A = \eta^3\frac{f_\Phi}{f_r} = \mu_y\mu_r, \qquad \mu_r = \eta^2. \tag{226}$$

We next show that

$$\mu_r \le \mu_y. \tag{227}$$

For this purpose, we need a simple estimate on r_{Φ_p}. Since

$$r_{\Phi_p}^2 = \eta w_1^2 + \left(p+\eta w_3\right)^2, \tag{228}$$

a direct computation gives

$$\left(p+\eta r\right)^2 - r_{\Phi_p}^2 = p\eta\left(r-w_3\right)\left(2-r-w_3\right) \ge 0. \tag{229}$$

Thus

$$r_{\Phi_p} \le p + \eta r. \tag{230}$$

Moreover,

$$\frac{d}{dx}\log f(x) = \frac{1}{(1-x^2)\operatorname{arctanh} x} - \frac{1}{x} - \frac{2x}{1-x^2}. \tag{231}$$

Therefore

$$\frac{d}{dx}\log f(x) + \frac{1}{1-x} = \frac{1}{(1-x^2)\operatorname{arctanh} x} - \frac{1}{x} + \frac{1}{1+x} \qquad = \frac{1}{(1-x^2)\operatorname{arctanh} x} - \frac{1}{x\,(1+x)}.$$

Thus it is enough to prove

$$(1-x)\operatorname{arctanh} x \le x. \tag{232}$$

This follows from

$$\operatorname{arctanh} x = \int_0^x \frac{dt}{1-t^2} \le \int_0^x \frac{dt}{1-x} = \frac{x}{1-x}. \tag{233}$$

Hence

$$\frac{d}{dx}\log f(x) \ge -\frac{1}{1-x}. \tag{234}$$

Integrating (234) from r to $p+\eta r$ gives

$$f\left(p+\eta r\right) \ge \eta f(r). \tag{235}$$

Since f is decreasing and $r_{\Phi_p} \le p+\eta r$, we obtain

$$f_\Phi = f\left(r_{\Phi_p}\right) \ge f\left(p+\eta r\right) \ge \eta f_r. \tag{236}$$

Consequently,

$$\mu_y = \eta\frac{f_\Phi}{f_r} \ge \eta^2 = \mu_r. \tag{237}$$

It remains to prove $\mu_+ \le \mu_y$. Define

$$n(x) = \frac{1-f(x)}{x^2}, \qquad 0 < x < 1, \tag{238}$$

with the continuous extension $n(0) = \frac{2}{3}$. Since

$$f(x) = \left(1-x^2\right)\frac{\operatorname{arctanh} x}{x}, \tag{239}$$

we have

$$n(x) = \frac{1 - \left(1 - x^2\right) \dfrac{\operatorname{arctanh} x}{x}}{x^2} = \int_0^1 \frac{1 - t^2}{1 - x^2 t^2}, dt. \tag{240}$$

In particular, $0 < n(x) < 1$, and n is increasing on $[0, 1)$.

The following scalar inequality is the key point:

$$n\left(r_{\Phi_p}\right) \leq \frac{n(r)}{\eta + p n(r)}. \tag{241}$$

To prove it, define

$$\ell(x) = \frac{1}{n(x)} - 1 = \frac{1 - n(x)}{n(x)}. \tag{242}$$

Since n is increasing and $0 < n < 1$, the function ℓ is positive and decreasing.

We claim that

$$\frac{d}{dx} \log \ell(x) \geq -\frac{1}{1 - x}. \tag{243}$$

Indeed, from (240), let

$$\phi_x(t) = \frac{1 - t^2}{1 - x^2 t^2}, \qquad \psi_x(t) = \frac{\left(1 - x^2\right) t^2}{1 - x^2 t^2}. \tag{244}$$

Then

$$\phi_x(t) + \psi_x(t) = 1, \tag{245}$$

and

$$n(x) = \int_0^1 \phi_x(t) dt, \quad 1 - n(x) = \int_0^1 \psi_x(t) dt. \tag{246}$$

Moreover,

$$n'(x) = \int_0^1 \frac{2x t^2 \left(1 - t^2\right)}{\left(1 - x^2 t^2\right)^2} dt = \frac{2x}{1 - x^2} \int_0^1 \phi_x(t) \psi_x(t) dt. \tag{247}$$

The function $\phi_x(t)$ is decreasing in t, while $\psi_x(t)$ is increasing in t. Hence, by the reverse Chebyshev inequality,

$$\int_0^1 \phi_x(t) \psi_x(t), dt \leq \left(\int_0^1 \phi_x(t), dt \right) \left(\int_0^1 \psi_x(t), dt \right) = n(x) \left(1 - n(x)\right). \tag{248}$$

Therefore

$$n'(x) \leq \frac{2x}{1-x^2} n(x)\,(1-n(x)) \leq \frac{1}{1-x} n(x)\,(1-n(x))\,, \tag{249}$$

where we used $\frac{2x}{1+x} \leq 1$ for $0 \leq x < 1$. Hence

$$\frac{d}{dx} \log \ell(x) = -\frac{n'(x)}{n(x)\,(1-n(x))} \geq -\frac{1}{1-x}, \tag{250}$$

which proves (243).

Integrating (243) from r to $p + \eta r$, we get

$$\ell\,(p+\eta r) \geq \eta \ell(r). \tag{251}$$

Since ℓ is decreasing and $r_{\Phi_p} \leq p + \eta r$, it follows that

$$\ell\left(r_{\Phi_p}\right) \geq \ell\,(p+\eta r) \geq \eta \ell(r). \tag{252}$$

This is equivalent to

$$\frac{1}{n\left(r_{\Phi_p}\right)} - 1 \geq \eta \left(\frac{1}{n(r)} - 1\right), \tag{253}$$

or

$$\frac{1}{n\left(r_{\Phi_p}\right)} \geq \frac{\eta}{n(r)} + p. \tag{254}$$

Thus

$$n\left(r_{\Phi_p}\right) \leq \frac{n(r)}{\eta + p n(r)}. \tag{255}$$

This proves (241).

We now finish the comparison. Let

$$n_r = n(r), \qquad n_\Phi = n\left(r_{\Phi_p}\right). \tag{256}$$

A direct expansion of $\mu_y + \mu_r - \operatorname{tr} A$ gives

$$\mu_y + \mu_r - \operatorname{tr} A = \frac{p\eta w_1^2}{f_r} \left[n_r - (\eta + p n_r)\, n_\Phi\right]. \tag{257}$$

For completeness, we indicate the algebra leading to this factorization. First one obtains

$$\mu_y + \mu_r - \operatorname{tr} A = \frac{p\eta w_1^2}{f_r r^2 r_{\Phi_p}^2} \Big[\eta r^2\,(f_\Phi - f_r) + p\,(1-f_r)\left(f_\Phi - \eta\,(1-w_3)^2\right)\Big]. \tag{258}$$

Using

$$1 - f_r = r^2 n_r, \qquad 1 - f_\Phi = r_{\Phi_p}^2 n_\Phi, \tag{259}$$

together with the identity

$$1 - r_{\Phi_p}^2 = \eta \left[1 - r^2 + p \left(1 - w_3 \right)^2 \right], \tag{260}$$

the last expression reduces exactly to (257).

By (241), we have

$$n_r - \left(\eta + p n_r \right) n_\Phi \geq 0. \tag{261}$$

Therefore

$$\mu_y + \mu_r - \operatorname{tr} A \geq 0. \tag{262}$$

The characteristic polynomial of A is

$$\chi_A(\mu) = \mu^2 - (\operatorname{tr} A)\, \mu + \det A. \tag{263}$$

Using (226), we get

$$\chi_A \left(\mu_y \right) = \mu_y^2 - (\operatorname{tr} A)\, \mu_y + \mu_y \mu_r = \mu_y \left(\mu_y + \mu_r - \operatorname{tr} A \right) \geq 0. \tag{264}$$

Since A is positive definite, its eigenvalues satisfy

$$0 < \mu_- \leq \mu_+. \tag{265}$$

Thus

$$\chi_A(\mu) = \left(\mu - \mu_- \right) \left(\mu - \mu_+ \right). \tag{266}$$

The inequality $\chi_A \left(\mu_y \right) \geq 0$ implies either

$$\mu_y \leq \mu_- \qquad \text{or} \qquad \mu_y \geq \mu_+. \tag{267}$$

The first possibility cannot occur strictly. Indeed, by (227),

$$\mu_- \mu_+ = \det A = \mu_y \mu_r \leq \mu_y^2. \tag{268}$$

If $\mu_y < \mu_-$, then

$$\mu_- \mu_+ \geq \mu_-^2 > \mu_y^2, \tag{269}$$

which is a contradiction. Hence

$$\mu_+ \leq \mu_y. \tag{270}$$

Therefore,

$$\lambda_{\pm} = \frac{1-r^2}{1-r_{\Phi_p}^2}\mu_{\pm} \leq \frac{1-r^2}{1-r_{\Phi_p}^2}\mu_y = \lambda_y. \tag{271}$$

Combining this with (216), we obtain

$$\lambda_{\max}\left(M^{-1}N\right) = \lambda_y = \eta\frac{1-r^2}{1-r_{\Phi_p}^2}\frac{f_\Phi}{f_r}. \tag{272}$$

Equivalently,

$$\lambda_{\max}\left(M^{-1}N\right) = (1-p)\,\frac{1-|\vec{w}|^2}{1-|\vec{w}_{\Phi_p}|^2}\frac{f\left(|\vec{w}_{\Phi_p}|\right)}{f\left(|\vec{w}|\right)}. \tag{273}$$

This is the desired spectral comparison.

E Computation of the eigenvalue infimum

Proof. For $\vec{v} = (y_1, 0, y_3)$, let

$$r = |\vec{v}| = \sqrt{y_1^2 + y_3^2}. \tag{274}$$

Define the two-dimensional matrix

$$\mathcal{M}(\vec{v}) = \begin{pmatrix} \dfrac{f(r)y_3^2 + y_1^2}{r^2} & \dfrac{\big(1-f(r)\big)y_1y_3}{r^2} \\ \dfrac{\big(1-f(r)\big)y_1y_3}{r^2} & \dfrac{f(r)y_1^2 + y_3^2}{r^2} \end{pmatrix}, \tag{275}$$

where at $r = 0$ the matrix is understood by continuous extension, namely

$$\mathcal{M}(0) = I_2. \tag{276}$$

Let

$$D_q = \operatorname{diag}\left(\sqrt{1-q}, 1-q\right). \tag{277}$$

Then the two matrices which determine the above quadratic equation are

$$\mathcal{M}(\vec{w}), \qquad D_q\mathcal{M}(\vec{w}_{\Phi_q})D_q. \tag{278}$$

Hence the number

$$\mu_-(\vec{w}) = \frac{C - \sqrt{C^2 - 4(1-q)^3 f(|\vec{w}|)f(|\vec{w}_{\Phi_q}|)}}{2f(|\vec{w}|)} \tag{279}$$

is the smaller generalized eigenvalue of the pair

$$\mathcal{M}(\vec{w}), \qquad D_q\mathcal{M}(\vec{w}_{\Phi_q})D_q. \tag{280}$$

Equivalently,

$$\mu_-(\vec{w}) = \min_{u\neq 0} \frac{u^T D_q\mathcal{M}(\vec{w}_{\Phi_q})D_q u}{u^T\mathcal{M}(\vec{w})u}. \tag{281}$$

Therefore

$$\lambda_-(\vec{w}) = \frac{1-|\vec{w}|^2}{1-|\vec{w}_{\Phi_q}|^2} \min_{u\neq 0} \frac{u^T D_q\mathcal{M}(\vec{w}_{\Phi_q})D_q u}{u^T\mathcal{M}(\vec{w})u}. \tag{282}$$

We first prove that

$$D_q\mathcal{M}(\vec{w}_{\Phi_q})D_q \geq (1-q)^2\mathcal{M}(\vec{w}). \tag{283}$$

Put

$$\eta = 1-q, \qquad \tau = -\log\eta. \tag{284}$$

Define

$$\vec{W}_\tau = \left(e^{-\frac{\tau}{2}}w_1, 0, 1-e^{-\tau}+e^{-\tau}w_3\right). \tag{285}$$

Then $\vec{W}_\tau = \vec{w}_{\Phi_q}$ when $e^{-\tau} = 1-q$. Let

$$P_\tau = \mathrm{diag}\left(e^{\frac{\tau}{2}}, 1\right). \tag{286}$$

Since

$$P_\tau = e^\tau D_q, \tag{287}$$

it is enough to prove that

$$H(\tau) := P_\tau\mathcal{M}(\vec{W}_\tau)P_\tau \tag{288}$$

is increasing in τ.

Write

$$\vec{W}_\tau = (rS, 0, rZ), \qquad S^2+Z^2 = 1, \qquad F = f(r). \tag{289}$$

In the orthonormal basis

$$n = (S, Z), \qquad m = (Z, -S), \tag{290}$$

the matrix $\mathcal{M}(\vec{W}_\tau)$ is diagonal:

$$\mathcal{M}(\vec{W}_\tau) = \begin{pmatrix} 1 & 0 \\ 0 & F \end{pmatrix}. \tag{291}$$

Using

$$rf'(r) = 1 - \frac{1+r^2}{1-r^2} f(r), \tag{292}$$

a direct computation gives

$$P_\tau^{-1} H'(\tau) P_\tau^{-1} = K, \tag{293}$$

where, in the basis $\{n, m\}$,

$$K = \begin{pmatrix} S^2 & S\left(Z - \dfrac{1-F}{r}\right) \\ S\left(Z - \dfrac{1-F}{r}\right) & f'(r)\left(Z - \dfrac{r}{2}(1+Z^2)\right) + FZ^2 \end{pmatrix}. \tag{294}$$

Thus it remains to prove $K \geq 0$. Let

$$B_r(Z) = f'(r)\left(Z - \frac{r}{2}(1+Z^2)\right) + FZ^2 - \left(Z - \frac{1-F}{r}\right)^2. \tag{295}$$

Then

$$\det K = S^2 B_r(Z). \tag{296}$$

Hence it is enough to prove

$$B_r(Z) \geq 0, \qquad -1 \leq Z \leq 1. \tag{297}$$

Since

$$\frac{f(r)}{1-r^2} = \frac{\operatorname{arctanh} r}{r} = \int_0^1 \frac{ds}{1 - r^2 s^2}, \tag{298}$$

Jensen's inequality gives

$$\frac{\operatorname{arctanh} r}{r} > \frac{1}{1 - \frac{r^2}{3}} = \frac{3}{3 - r^2}. \tag{299}$$

Therefore

$$(3 - r^2) f(r) > 3(1 - r^2). \tag{300}$$

The coefficient of Z^2 in $B_r(Z)$ is

$$\alpha_r = \frac{(3-r^2)f(r) - 3(1-r^2)}{2(1-r^2)}, \tag{301}$$

and hence $\alpha_r > 0$. Moreover, direct simplification gives

$$B_r'(Z) = 2\alpha_r \left(Z - \frac{1}{r} \right). \tag{302}$$

Since $0 < r < 1$, one has $\frac{1}{r} > 1$. Thus $B_r'(Z) < 0$ on $[-1, 1]$, and consequently

$$B_r(Z) \geq B_r(1). \tag{303}$$

It remains to prove $B_r(1) > 0$. Let

$$L = \text{arctanh}r. \tag{304}$$

A direct computation gives

$$B_r(1) = \frac{1-r}{r^4} J(r), \tag{305}$$

where

$$J(r) = (r^3 + r^2 - r - 1)L^2 - (r^3 + r^2 - 2r)L + 2r^3 - r^2. \tag{306}$$

Furthermore,

$$J^{(4)}(r) = \frac{16\left(2(1+r)^2 L + r^2 - 2r\right)}{(1-r)^3(1+r)^4}. \tag{307}$$

Since $L = \text{arctanh}r > r$, we have

$$2(1+r)^2 L + r^2 - 2r > 2(1+r)^2 r + r^2 - 2r = 5r^2 + 2r^3 > 0. \tag{308}$$

Hence

$$J^{(4)}(r) > 0, \qquad 0 < r < 1. \tag{309}$$

Since

$$J(0) = J'(0) = J''(0) = J'''(0) = 0, \tag{310}$$

we obtain

$$J(r) > 0, \qquad 0 < r < 1. \tag{311}$$

Therefore

$$B_r(1) > 0, \tag{312}$$

and hence

$$B_r(Z) \geq 0, \qquad -1 \leq Z \leq 1. \tag{313}$$

This proves $K \geq 0$, so $H'(\tau) \geq 0$. Consequently,

$$H(\tau) \geq H(0) = \mathcal{M}(\vec{w}). \tag{314}$$

Since $e^{-\tau} = 1 - q$, this implies

$$D_q \mathcal{M}(\vec{w}_{\Phi_q}) D_q \geq (1-q)^2 \mathcal{M}(\vec{w}). \tag{315}$$

Therefore

$$\min_{u \neq 0} \frac{u^T D_q \mathcal{M}(\vec{w}_{\Phi_q}) D_q u}{u^T \mathcal{M}(\vec{w}) u} \geq (1-q)^2. \tag{316}$$

Next we estimate the scalar factor. Parametrize A by

$$\vec{w} = T_{p_1}(x,z) := \left(\sqrt{1-p_1}x, 0, p_1 + (1-p_1)z\right), \qquad x^2 + z^2 \leq 1. \tag{317}$$

Then

$$\vec{w}_{\Phi_q} = T_{\widehat{q}}(x,z), \tag{318}$$

because

$$\widehat{q} = q + p_1 - q p_1. \tag{319}$$

For $0 < s < 1$, we have

$$1 - |T_s(x,z)|^2 = (1-s)(R + sd^2), \tag{320}$$

where

$$R = 1 - x^2 - z^2, \qquad d = 1 - z. \tag{321}$$

Thus

$$\frac{1 - |\vec{w}|^2}{1 - |\vec{w}_{\Phi_q}|^2} = \frac{(1-p_1)(R + p_1 d^2)}{(1-\widehat{q})(R + \widehat{q} d^2)}. \tag{322}$$

Since

$$1 - \widehat{q} = (1-q)(1-p_1), \tag{323}$$

we get

$$\frac{1-|\vec{w}|^2}{1-|\vec{w}_{\Phi_q}|^2} = \frac{1}{1-q}\frac{R+p_1 d^2}{R+\widehat{q}d^2}. \tag{324}$$

Because $\widehat{q} > p_1$ and $R \geq 0$, one has

$$\frac{R+p_1 d^2}{R+\widehat{q}d^2} \geq \frac{p_1}{\widehat{q}}. \tag{325}$$

Hence

$$\frac{1-|\vec{w}|^2}{1-|\vec{w}_{\Phi_q}|^2} \geq \frac{p_1}{(1-q)\widehat{q}}. \tag{326}$$

Combining the two estimates gives

$$\lambda_-(\vec{w}) \geq \frac{p_1}{(1-q)\widehat{q}}(1-q)^2 \tag{327}$$

$$= \frac{p_1(1-q)}{\widehat{q}}. \tag{328}$$

Therefore

$$h(q) = \inf_{\vec{w}\in A} \lambda_-(\vec{w}) \geq \frac{p_1(1-q)}{\widehat{q}}. \tag{329}$$

It remains to verify that equality can be attained. Take

$$x = 0, \qquad z = -1. \tag{330}$$

Then

$$\vec{w}_* = T_{p_1}(0,-1) = (0,0,2p_1-1), \tag{331}$$

and

$$(\vec{w}_*)_{\Phi_q} = T_{\widehat{q}}(0,-1) = (0,0,2\widehat{q}-1). \tag{332}$$

At this point,

$$\frac{1-|\vec{w}_*|^2}{1-|(\vec{w}_*)_{\Phi_q}|^2} = \frac{1-(2p_1-1)^2}{1-(2\widehat{q}-1)^2} \tag{333}$$

$$= \frac{4p_1(1-p_1)}{4\widehat{q}(1-\widehat{q})} \tag{334}$$

$$= \frac{p_1}{(1-q)\widehat{q}}. \tag{335}$$

On the other hand, when $w_1 = 0$, the matrices

$$\mathcal{M}(\vec{w}_*), \qquad D_q \mathcal{M}((\vec{w}_*)_{\Phi_q}) D_q \tag{336}$$

are simultaneously diagonalized, and the generalized eigenvalue in the w_3-direction is

$$(1-q)^2. \tag{337}$$

Hence

$$\lambda_-(\vec{w}_*) = \frac{p_1}{(1-q)\widehat{q}}(1-q)^2 \tag{338}$$

$$= \frac{p_1(1-q)}{\widehat{q}}. \tag{339}$$

Therefore the lower bound is attained, and we conclude that

$$h(q) = \min_{\vec{w} \in A} \lambda_-(\vec{w}) = \frac{p_1(1-q)}{\widehat{q}} = \frac{p_1(1-q)}{q + p_1 - q p_1}. \tag{340}$$

The minimum is attained at

$$\vec{w}_* = (0, 0, 2p_1 - 1). \tag{341}$$

□

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